Extremizers for the independent Berry–Esseen inequality

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Abstract

If the independent Berry–Esseen constant exceeds Esseen's constant, its supremum is attained in a class of countable independent convolutions with an infinitely divisible remainder. Every original coordinate of an unrestricted probability-local maximizer has at most four support points. At a full maximum one can choose at most three points per coordinate while preserving each coordinate's variance and absolute third moment. For a maximizing representation with these three-point coordinates, the Gaussian and Lévy components vanish, its summand supports shrink to zero, and an infinitely active sum is nonatomic. For these three-point representations, under the additional bound $\beta \le 18481/25000$ on the additive third-moment quantity at total variance one, we prove weighted-variance summability with exponent below $\sqrt{5}-1$, or below $\sqrt{21}-3$ for an eventually two-point representation. These estimates exclude specified regularity at a maximizing threshold. Contact variations also exclude separated finite blocks and Bernoulli tails satisfying explicit gap and nonoverlap conditions. The sharp inequality for arbitrary independent arrays remains open.

1 Statements and notation

For a centered random variable S of variance one, set

$$\Delta(S) = \sup_{x\in\mathbb{R}} |\mathbb{P}(S \le x) - \Phi(x)|.$$

The standard normal density and distribution function are denoted by ϕ and Φ. Define

$$C_{\mathrm{ind}} = \sup \frac{\Delta(\sum_{i=1}^n X_i)}{\sum_{i=1}^n \mathbb{E}|X_i|^3}, \qquad C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}},$$

where the supremum is over finite independent centered arrays with total variance one and finite third absolute moments. The classical Berry–Esseen theorem gives $C_{\mathrm{ind}} < \infty$, and Esseen's two-point asymptotics give $C_{\mathrm{ind}} \ge C_{\mathrm{E}}$ [6, 16].

We use the small-Lyapunov theorem proved in the companion manuscript [3, Theorem 1.1]: with $\ell_0 = \exp(-\exp(97))$,

$$\sum_i \mathbb{E}|X_i|^3 \le \ell_0 \quad \Longrightarrow \quad \Delta(\textstyle\sum_i X_i) \le C_{\mathrm{E}} \sum_i \mathbb{E}|X_i|^3. \tag{1.1}$$

That theorem extends the uniform smoothing, contamination and local comparison methods in the first version of He and Cheng [9] to nonidentically distributed arrays, with explicit constants. Its finite input is the common-diameter Bernoulli theorem of [2]. The present paper concerns the remaining extremal problem at positive Lyapunov ratio.

The support theorem below also applies to full maxima at a fixed finite horizon. Its probability variations preserve each coordinate's mass and mean, while allowing its variance and absolute third moment to change. The subsequent compactification, contact propagation and Gaussian exclusion concern the unrestricted horizon supremum. These are different comparison classes; fixing every individual variance does not supply the probability variations used for the support theorem.

Definition 1.1. An admissible representation is a decomposition of the form

$$S = \sum_{i\ge 1} X_i + Y,$$

$$\log \mathbb{E}e^{itY} = -\frac{at^2}{2} + \int_{\mathbb{R}\setminus\{0\}} (e^{ity} - 1 - ity)\,\nu(dy), \tag{1.2}$$

$$\sum_i \operatorname{Var}(X_i) + a + \int y^2\nu(dy) = 1, \qquad \beta = \sum_i \mathbb{E}|X_i|^3 + \int |y|^3\nu(dy) < \infty. \tag{1.3}$$

Here the X_i are independent centered variables, and Y is a centered infinitely divisible variable independent of all X_i. The number a is nonnegative, the series converges in L^2, and ν is a Lévy measure with finite second moment. The number β is the additive third-moment quantity of the representation. A representation is maximizing if $\beta > 0$ and $\Delta(S) = C_{\mathrm{ind}}\beta$.

The Gaussian term contributes zero to β. Indeed a Gaussian variable of variance a can be approximated by sums whose variance is a and whose third-moment sums tend to zero. Thus β is not the third absolute moment of S, and it is not determined by the law of S alone. The approximation statement for the whole representation is proved in Section 2.

Theorem 1.2. *If $C_{\mathrm{ind}} > C_{\mathrm{E}}$, there is a maximizing admissible representation. Every maximizing representation has at most four support points per original coordinate, and it admits an equally maximizing representation with at most three points per coordinate that preserves each original variance and absolute third moment, and the independent remainder. Every maximizing representation with at most three support points per coordinate satisfies*

$$a = 0, \qquad \nu = 0, \qquad \max\{|u| : u \in \operatorname{supp} X_i\} \longrightarrow 0. \tag{1.4}$$

If infinitely many X_i are nonconstant, then the law of S is nonatomic and

$$\sum_i (1 - \max_u \mathbb{P}(X_i = u)) = \infty.$$

Reflection, if needed, gives a threshold z at which

$$F(z) - \Phi(z) = D = C\beta > 0, \qquad C = C_{\mathrm{ind}}, \qquad A = \frac{3D + z\phi(z)}{2}, \qquad F(x) = \mathbb{P}(S \le x). \tag{1.5}$$

All CDFs in this paper are closed CDFs. A left limit is written explicitly. The function $F - \Phi$ is upper semicontinuous and tends to zero at both ends of the line, so a positive supremum on this branch is attained. The continuous normal CDF allows reflection to interchange the positive branch with the branch $\Phi(x) - \mathbb{P}(S < x)$.

Theorem 1.3 (Weighted variance of the original coordinates)**.** *Suppose $C_{\mathrm{ind}} > C_{\mathrm{E}}$, and let a maximizing representation with at most three support points per original coordinate have $\beta \le 18481/25000$. For each nonconstant coordinate put $H_i = \operatorname{diam}(\operatorname{supp} X_i)$ and $v_i = \operatorname{Var}(X_i)$. Then*

$$\sum_i \frac{v_i}{H_i^\gamma} < \infty \quad (0 < \gamma < \sqrt{5} - 1), \qquad \sum_i \mathbb{E}|X_i| < \infty.$$

If all but finitely many nonconstant coordinates have two support points, the weighted-variance range improves to $0 < \gamma < \sqrt{21} - 3$. *These statements concern the original independent coordinates.*

Corollary 1.4 (Regularity excluded at a maximizing threshold)**.** *Under the hypotheses of Theorem 1.3, a threshold satisfying* (1.5) *cannot obey*

$$F(z+t) = F(z) + \phi(z)t + O(t^2) \qquad (t \to 0)$$

through both signs. If all but finitely many nonconstant coordinates have two support points, it cannot obey

$$F(z) - F(z-h) = \phi(z)h + O(h^{1+\alpha}) \qquad (h \downarrow 0)$$

for any $\alpha > (\sqrt{21} - 3)/6$.

The small-Lyapunov input prevents the third-moment quantity of a maximizing sequence from tending to zero. Compactification supplies the full relaxed comparison class. The probability Hessian then gives Theorem 3.2, including its exact moment-preserving three-point representative. Contact-copy variations remove the Lévy remainder and nonshrinking supports; a separate lattice and Fourier comparison removes the Gaussian remainder. Together these give Theorem 1.2. For the quantitative restrictions, the additional bound on β controls the shapes of three-point supports. Grouping small coordinates and comparing with Gaussian and lattice sums give weighted summability. The grouping rates also bound how closely the distribution function can follow its tangent at a maximizing threshold; the rare-tail exclusions apply this constraint to specified sequences of gaps.

The third-moment bound supplies the two strict curvature inequalities in Lemma 8.1; it is not a consequence of attainment. Write $W_s = \sum_i v_i/H_i^s$, summing over nonconstant coordinates. The proof first establishes $W_s < \infty$ for some $s > 1$. Bounds on the number of coordinates above a variance threshold then permit more efficient grouping of the remaining coordinates. Gaussian and lattice comparisons turn the grouping estimates into $W_\gamma < \infty$ for every $0 < \gamma < (3s+4)/(s+5)$. Iteration reaches every exponent below the positive fixed point $\sqrt{5} - 1$. When all but finitely many nonconstant coordinates have two support points, grouping by the sign of the nonmodal displacement gives the map $(3s+12)/(s+9)$, whose positive fixed point is $\sqrt{21} - 3$. Sections 8–12 give the full argument.

There is also a restriction on support configurations. For a finite set I of indices, put $T_I = S - \sum_{i \in I} X_i$ and

$$\mathcal{A}_I = \left\{ (x_i)_{i \in I} \in \prod_{i \in I} \operatorname{supp} X_i : z - \sum_{i \in I} x_i \in \operatorname{supp} T_I \right\}. \tag{1.6}$$

The support on the right is the closed support of the actual independent remainder. This definition does not condition on the event $S = z$.

Theorem 1.5. *Fix a maximizing representation with at most three support points per original coordinate, as in Theorem 1.2, and a threshold satisfying* (1.5)*. There is* k_0 *such that, for every finite* $I \subset \{k_0, k_0+1, \dots\}$*, the following holds: if functions* $f_i : \operatorname{supp} X_i \to \mathbb{R}$ *satisfy*

$$\sum_{i \in I} f_i(x_i) = c \quad \text{for all } (x_i) \in \mathcal{A}_I,$$

then every f_i *is constant. Equivalently, choose one state* $u_i^0 \in \operatorname{supp} X_i$ *for each* i*. The indicator vectors* $\left(\mathbf{1}_{\{x_i = u\}}\right)_{i \in I,\, u \in \operatorname{supp} X_i \setminus \{u_i^0\}}$ *associated with* $\mathcal{A}_I$ *have full affine dimension* $\sum_{i \in I} (|\operatorname{supp} X_i| - 1)$.

Corollary 1.6. *Such a representation cannot consist of a finite-support independent prefix followed by infinitely many nonconstant finite blocks of finite-support summands for which*

$$H_k = \sum_{i \text{ in block } k} \operatorname{diam}(\operatorname{supp} X_i), \qquad \delta_k = \min_{u \neq v \in \operatorname{supp}(\sum_{i \text{ in block } k} X_i)} |u - v|$$

satisfy

$$\sum_k H_k < \infty, \qquad \delta_k \geq \sum_{l>k} H_l. \tag{1.7}$$

Theorem 1.2 leaves both finite representations and nonatomic infinite convolutions with shrinking supports. Such infinite convolutions need not have a density. Theorem 1.5 is automatic when every relevant remainder has support $\mathbb{R}$. Under the additional third-moment bound, Theorem 1.3 gives weighted summability, and Corollary 1.4 excludes the stated local expansions of the distribution function.

The compactification uses the classical accompanying Poisson-law method and Lévy–Khintchine representation [13]; Section 2 proves the needed moment bounds. Coordinate contamination follows He and Cheng [9, Lemma 4.3]; the independent-coordinate derivatives, Lévy-intensity variations and contact propagation are derived below. The Gaussian estimates use the zero-bias transformation of Goldstein and Reinert [8], the transport estimate independently obtained by Goldstein and Tyurin [7, 18], and the compound-Poisson Berry–Esseen bound of Korolev and Shevtsova [12]. The atom argument uses Sperner's theorem [17]. These inputs are stated where they enter the proofs.

2 Compactification and attainment

Lemma 2.1. *For every $c > 0$, a finite independent variance-one array satisfying $\Delta(\sum_i X_i) > c \sum_i \mathbb{E}|X_i|^3$ can be replaced by one with the same number and variances of summands, each having at most three support points, for which the same strict inequality holds.*

Proof. Choose the positive closed-CDF branch by reflection. At its fixed threshold, with all other summands held fixed, maximize

$$\int \{G_i(z-u) - c|u|^3\}\, P(du) \quad \text{subject to} \int u\, dP = 0, \quad \int u^2\, dP = v_i.$$

For $v_i = 0$ the law is fixed. Otherwise the objective is upper semicontinuous and bounded above by $1 - c|u|^3$. A maximizing sequence has bounded third moments, so a weak limit preserves its first two moments and attains the supremum. The maximizing set is compact and is a face of the convex moment set. It has an extreme point. Such a point has at most three atoms: four disjoint positive-mass sets would give a linear dependence among their mass, first-moment and second-moment vectors, and hence two distinct feasible bounded-density perturbations with the original law as their average. This is the elementary three-constraint case of the moment-set principle in Winkler [19]; the coercive argument is also given in Bell [2]. Replacing coordinates successively preserves the strict positive excess at coefficient c. □

Lemma 2.2. *Let finite independent centered arrays have total variance one and third-moment sums $\beta_n \leq M$. After passage to a subsequence, their sums converge weakly to an admissible representation with*

$$\beta \leq \liminf_n \beta_n.$$

If the original summands have at most three atoms, so do the limiting independent variables X_i.

Proof. First take a subsequence along which β_n tends to its liminf. Order each row by decreasing variance, padding it with zero variables. Then $v_{n,k} \le 1/k$. The laws are tight, and

$$\mathbb{E}[X_{n,k}^2; |X_{n,k}| > L] \le M/L$$

makes their squares uniformly integrable. A diagonal subsequence gives $X_{n,k} \Rightarrow X_k$, convergence of means and variances, and convergence of the numbers $\mathbb{E}|X_{n,k}|^3$ to numbers b_k. Writing $v_k = \mathbb{E}X_k^2$ and $r_k = \mathbb{E}|X_k|^3$, we have

$$\sum_k v_k \le 1, \qquad r_k \le b_k, \qquad \sum_k b_k \le \lim_n \beta_n.$$

The independent series $\sum_k X_k$ exists in L^2.

Choose $K_n \to \infty$ sufficiently slowly that the first K_n laws approximate their limits on each fixed prefix and $\sum_{k \le K_n} |v_{n,k} - v_k| \to 0$. The corresponding prefix sums converge weakly to $\sum_k X_k$: after fixing a finite prefix, the variance of what remains in the growing prefix is at most $\sum_{k>K} v_k + o(1)$. Define the quadratic measures of the remaining row by

$$\Lambda_n(E) = \sum_{k > K_n} \mathbb{E}[X_{n,k}^2 \mathbf{1}_{\{X_{n,k} \in E\}}].$$

Their masses tend to $v_0 = 1 - \sum_k v_k$ and their tails are bounded by M/L outside $[-L, L]$. Take $\Lambda_n \Rightarrow \Lambda$; its mass is v_0.

For a centered variable of variance v, its characteristic function f satisfies $|f(t) - 1| \le vt^2/2$. If $\Re w \le 0$, then $|e^w - 1 - w| \le |w|^2/2$. Telescoping products therefore gives

$$\left| \prod_j f_j(t) - \exp\Big\{ \sum_j (f_j(t) - 1) \Big\} \right| \le \frac{t^4}{8} \sum_j v_j^2. \tag{2.1}$$

For the remaining row the right side tends to zero, since its largest variance is at most $1/(K_n + 1)$. Its accompanying Poisson exponent is

$$\int q_t(y)\, \Lambda_n(dy), \qquad q_t(y) = \frac{e^{ity} - 1 - ity}{y^2}, \quad q_t(0) = -t^2/2.$$

The function q_t is bounded and continuous. Hence these exponents converge to the integral against Λ. Set $a = \Lambda(\{0\})$ and $\nu(dy) = y^{-2}\Lambda(dy)$ off zero. This gives (1.2), with variance v_0. Independence of the original prefix and tail proves the claimed representation and preserves total variance.

For each fixed K and all sufficiently large n,

$$\int |y|\, \Lambda_n(dy) \le \beta_n - \sum_{k \le K} \mathbb{E}|X_{n,k}|^3.$$

Lower semicontinuity and then $K \to \infty$ give $\int |y|^3 \nu(dy) \le \lim \beta_n - \sum_k b_k$. Together with $r_k \le b_k$ this proves the third-moment inequality. No uniform integrability of cubic moments was assumed. Finally, a weak limit of three-atom laws cannot charge four disjoint open sets. □

Lemma 2.3. *Every admissible representation is a weak limit of finite centered independent arrays with variance exactly one and third-moment sums tending to its number β. Consequently*

$$\Delta(S) \le C_{\mathrm{ind}} \beta. \tag{2.2}$$

Proof. Keep finitely many X_i. Truncate ν away from zero and infinity, and approximate it by a finite atomic measure with convergent second and third absolute integrals. A small contraction can make its variance no larger than the available variance. Replace each atom $\lambda\delta_y$ by N independent variables $y(B-\lambda/N)$, where $B\sim\mathrm{Bern}(\lambda/N)$. Their total variance and third-moment sum are

$$\lambda y^2(1-\lambda/N),\qquad \lambda|y|^3(1-\lambda/N)\{(\lambda/N)^2+(1-\lambda/N)^2\}.$$

They converge to the corresponding compensated Poisson law and its second and third jump integrals. Fill the remaining variance by N independent symmetric two-point variables. If that variance is b, their third-moment sum is $b^{3/2}/\sqrt{N}$. Sending the truncation and approximation indices to their limits gives the required Gaussian variance and discards a tail of $\sum_i X_i$ whose variance tends to zero. Every finite array has variance one.

If $P_n\Rightarrow P$ and H is a continuous CDF, then $\|F_{P_n}-H\|_\infty\to\|F_P-H\|_\infty$. To verify this assertion, sandwich CDF values between a finite grid of continuity points of F_P on which the oscillation of H is arbitrarily small, and control both tails. Values immediately before and after jumps are recovered by continuity points and the continuity of H. Applying this to $H=\Phi$ proves (2.2). □

Proposition 2.4. *If $C_{\mathrm{ind}}>C_{\mathrm{E}}$, a maximizing representation with three-point independent variables exists.*

Proof. Take coefficients $c_n\uparrow C_{\mathrm{ind}}$ with $c_n>C_{\mathrm{E}}$ and finite arrays whose ratios exceed c_n. Lemma 2.1 supplies three-point arrays with the same property. Their third-moment sums satisfy $\ell_0<\beta_n<1/c_n$ by (1.1) and $\Delta\le 1$. Take $\beta_n\to r>0$ and apply Lemma 2.2. Weak continuity of distance to the normal gives $\Delta(S)=C_{\mathrm{ind}}r>0$. The limiting representation satisfies

$$C_{\mathrm{ind}}r=\Delta(S)\le C_{\mathrm{ind}}\beta\le C_{\mathrm{ind}}r.$$

Thus $\beta=r$ and it is maximizing. This equality rules out loss of the additive third-moment quantity in the compactification. □

3 Variations and contact identities

Fix a maximizing representation with $C=C_{\mathrm{ind}}>C_{\mathrm{E}}$ and (1.5). For an admissible perturbation of raw independent laws, let m be the total mean, B the total central variance, R_* the sum of the recentered third absolute moments plus the Lévy third integral, and F_* the raw sum CDF. At a raw threshold w, put

$$J=B^{3/2}\left\{F_*(w)-\Phi\left(\frac{w-m}{\sqrt{B}}\right)\right\}-CR_*. \tag{3.1}$$

By Lemma 2.3, $J\le 0$ whenever the perturbation is admissible, and $J=0$ at the original representation and $w=z$. Finite probability variations make $F_*(z)$ a multiaffine polynomial, including at atoms. Since $|x|^3$ is C^2, their central third moments are twice differentiable. A two-sided probability parameter consequently has zero first derivative, and the Hessian is negative semidefinite on its admissible tangent cone when all first derivatives vanish.

Lemma 3.1. *Let G_i be the CDF of $S-X_i$, and put $v_i=\mathbb{E}X_i^2$, $\rho_i=\mathbb{E}|X_i|^3$, $M_i=\mathbb{E}[X_i|X_i|]$. Then*

$$G_i(z-u)-F(z)+\phi(z)u+A(u^2-v_i)-C\{|u|^3-\rho_i-3M_iu\}\le 0, \tag{3.2}$$

$$Q(y):=F(z-y)-F(z)+\phi(z)y+Ay^2-C|y|^3\le 0. \tag{3.3}$$

Equality holds in (3.2) at every support point of X_i, and $Q(y)=0$ at every nonzero point of $\operatorname{supp}\nu$. *All supports* $\operatorname{supp}X_i$ *lie in one bounded interval, and all nonzero zeros of Q lie in one bounded interval.*

Proof. Replace the raw law of X_i by $(1-t)\,\mathrm{Law}(X_i) + t\delta_u$. The derivatives of mean, variance and third absolute central moment are u, $u^2 - v_i$ and $|u|^3 - \rho_i - 3M_i u$. Differentiating (3.1) gives (3.2). Its integral under $\mathrm{Law}(X_i)$ is zero. Upper semicontinuity then extends equality from almost every point to the entire support. This is the coordinatewise contamination argument of He and Cheng [9, Lemma 4.3].

Append a compensated Poisson variable $y(N_t - t)$ and use the centered threshold $z - ty$. Its raw CDF is exactly $\mathbb{E}F(z - yN_t)$; variance and third-moment quantity increase by ty^2 and $t|y|^3$. Its right first derivative is $Q(y)$, proving (3.3). For a finite signed measure $\eta = f\nu$, with bounded f supported away from zero, both signs of small changes in ν are admissible. Their derivatives are $\int Q\, d\eta$. Thus $Q = 0$ almost everywhere for ν, and upper semicontinuity gives equality on its nonzero support.

In (3.2), use $v_i \le 1$, $|M_i| \le 1$ and $C\rho_i \le C\beta = D$. Its right side at a support point is bounded above by a fixed cubic polynomial in $|u|$ with leading coefficient $-C$. The same reasoning applies to (3.3). Both zero sets are therefore bounded uniformly in i. □

For later calculations, consider a mass transfer between two states u, v of one centered variable. Write $d = u - v$, $b = u^2 - v^2$, and let $g = G_i(z - u) - G_i(z - v)$. The third-moment derivatives are

$$\begin{aligned} r' &= |u|^3 - |v|^3 - 3d\mathbb{E}[X_i|X_i|], \\ r'' &= -6d(u|u| - v|v|) + 6d^2\mathbb{E}|X_i|. \end{aligned} \tag{3.4}$$

Stationarity says $g + \phi d + Ab - Cr' = 0$. Hereafter ϕ without an argument in an algebraic formula means $\phi(z)$. Put

$$B_0 = \frac{3D + (3z + z^3)\phi}{4}, \qquad E_0 = (1 + z^2/2)\phi.$$

For two transfers in distinct coordinates the Hessian entries of (3.1) are

$$\begin{aligned} H_{ii} &= 3b_i g_i + B_0 b_i^2 + 2E_0 b_i d_i - 3Dd_i^2 - Cr_i'', \\ H_{ij} &= g_{ij} + \tfrac{3}{2}(b_i g_j + b_j g_i) + B_0 b_i b_j + E_0(b_i d_j + b_j d_i) + z\phi d_i d_j, \quad i \ne j. \end{aligned} \tag{3.5}$$

Here g_{ij} is the exact raw double CDF difference. These identities follow by differentiating $B^{3/2}\Phi((z-m)/\sqrt{B})$ at $(B, m) = (1, 0)$; its second derivatives are

$$N_{mm} = -z\phi, \quad N_{mB} = -(1 + z^2/2)\phi, \quad N_{BB} = \tfrac{3}{4}\Phi(z) - \tfrac{1}{4}(3z + z^3)\phi.$$

The second variance derivative in one transfer is $-2d_i^2$, whereas mixed moment derivatives in different coordinates vanish. This accounts for the diagonal term $-3Dd_i^2$ in (3.5).

3.1 Support of every original coordinate

The following result has a larger finite-horizon domain than the maximizing representations considered elsewhere in this paper. A finite array is priced by $R = \sum_i \mathbb{E}|X_i|^3$. In the countable class, R also includes the Lévy cubic integral as in (1.3); its Gaussian remainder has zero price. For total variance $V > 0$ the objective is $V^{3/2} d_K(S, N(0, V))/R$.

Theorem 3.2 (Support of the original coordinates)**.** *Let independent centered originals, with finite summed variance and absolute third moments, form a finite array or the original block of an admissible representation. Suppose total variance V and total price R are positive. Select a positive closed-CDF branch $D = \mathbb{P}(S \le y) - \Phi(y/\sqrt{V}) > 0$, after reflection if needed. If its ratio $C = V^{3/2}D/R$ is locally maximal against all two-sided signed probability changes of one original*

with bounded density relative to its law and compact support, preserving its mass and mean, then every original has at most four support points.

If the full objective is globally maximal at a fixed finite horizon (zero coordinates allowed), or over the full admissible representation class, and the selected branch attains its full discrepancy, there is an equally maximizing representation with at most three support points per original. It preserves each original variance and absolute third moment, the selected threshold, and any independent remainder. In particular a four-point original at such a full maximum lies on an exact flat probability segment with a three-point endpoint.

Proof. Fix one original law P and a signed measure δ with bounded density relative to P, supported in a bounded set, and satisfying $\int d\delta = \int u\, d\delta = 0$. Both signs of sufficiently small ε make $P + \varepsilon\delta$ a centered probability law. Write

$$v_1 = \int u^2\, d\delta, \qquad r_1 = \int |u|^3\, d\delta, \qquad z = y/\sqrt{V}, \qquad q = z\phi(z)(3 - z^2).$$

All other components stay fixed. Thus total variance and original price are exactly $V + \varepsilon v_1$ and $R + \varepsilon r_1$. At the same raw closed event the CDF is affine:

$$F_\varepsilon(y) = \int \mathbb{P}(S - X \le y - u)\, (P + \varepsilon\delta)(du).$$

This requires no density of the deleted law and remains valid at atoms. Put $D_\varepsilon = F_\varepsilon(y) - \Phi(y/\sqrt{V + \varepsilon v_1})$ and $C_\varepsilon = (V + \varepsilon v_1)^{3/2} D_\varepsilon/(R + \varepsilon r_1)$. The two-sided local ceiling gives $C_0' = 0$, so

$$\frac{D_0'}{D} = \frac{r_1}{R} - \frac{3v_1}{2V}, \qquad D_0'' = -\frac{q v_1^2}{4V^2}.$$

Taking the stationary second logarithmic derivative yields

$$\frac{C_0''}{C} = -\frac{15 + q/D}{4}\left(\frac{v_1}{V}\right)^2 + \frac{3 r_1 v_1}{RV} \le 0. \tag{3.6}$$

Consequently the admissible directions cannot realize arbitrary pairs (v_1, r_1): choosing $v_1 = V$ and $r_1 = R\{(15 + q/D)/12 + 1\}$ would give $C_0''/C = 3$.

Any five distinct real points make the evaluation vectors $(1, u, u^2, |u|^3)$ have rank four. Indeed a nonzero relation with zero absolute-cubic coefficient is a quadratic and cannot vanish at five points. Otherwise a function $|u|^3 - a - bu - cu^2$ would have five zeros, although its continuous second derivative $6|u| - 2c$ has at most two zeros, contradicting Rolle's theorem twice. If P had five support points, choose four independent evaluation columns and disjoint bounded neighborhoods of their points with positive P mass. The normalized restrictions of P to sufficiently small fixed neighborhoods have an invertible four-moment matrix. A signed combination of these restrictions therefore realizes zero mass and mean and the forbidden (v_1, r_1). Its density relative to P is bounded: the neighborhood masses can be arbitrarily small but are positive and fixed. This supplies a nonempty two-sided probability interval and contradicts (3.6). Every actual support has at most four points, including in the countable class.

Suppose now that one original has four distinct atoms u_j of positive masses p_j. Its four-moment matrix has rank three: rank four would again realize the forbidden pair, while $(1, u, u^2)$ already has rank three. Hence $|u_j|^3 = a + bu_j + cu_j^2$ on this support. A nonzero vector (d_j) annihilating $(1, u, u^2)$ also annihilates $|u|^3$. Along the compact feasible interval

$$I = \{t : p_j + t d_j \ge 0 \text{ for all } j\},$$

mass, mean, variance and the original absolute third moment stay exactly fixed. The selected CDF is affine in t; stationarity at the interior point 0 makes its slope zero. At either endpoint at least one mass vanishes. Full globality makes that endpoint another global maximizer: its selected ratio equals C, and its full objective cannot exceed C.

Replace all four-point originals successively. This immediately proves the finite statement. For countably many originals couple old and new coordinates independently across indices. The difference between the partly replaced and fully replaced sums has variance at most $4\sum_{i>k} v_i \to 0$. Every individual price and variance has been preserved, so the limiting total V, R and residual are unchanged. Portmanteau at the same closed threshold gives $F_{\mathrm{new}}(y) \geq \limsup_k F_k(y) = F(y)$. The full relaxed ceiling forces equality. The fully replaced representation is therefore equally maximizing and retains the selected threshold. □

This theorem gives no positive atom-mass floor and no bound on the number of nonconstant originals. It does not assert that every representation has three-point coordinates. The three-point moment-face reduction used in compactification, Lemma 2.1, instead applies to near-supremizing arrays and need not preserve their individual cubic prices. No hypothesis $C_{\mathrm{ind}} > C_{\mathrm{E}}$ enters Theorem 3.2.

3.2 Original convolution factors and their price

The inequality below is the $p = 3$ lower bound in Cox–Kemperman's Theorem (2.6), with parameters (2.9); their equality condition also gives its strictness [4]. We give a proof using the centered two-point mixture reduction and mixed-derivative argument presented by Ruzankin [15, Theorem E and Sections 2.1–2.2]. The strict inequality will compare the third-moment costs of an original variable and its independent convolution factors.

Lemma 3.3. *For independent centered real U, W with finite absolute third moments,*

$$\mathbb{E}|U+W|^3 \geq \mathbb{E}|U|^3 + \mathbb{E}|W|^3.$$

Equality holds exactly when at least one variable is identically zero.

Proof. For centered two-point laws on $\{-a, b\}$ and $\{-c, d\}$ with all endpoint magnitudes positive, multiply the difference by $(a+b)(c+d)$ to obtain

$$\begin{aligned} N = {} & bd\{(a+c)^3 - a^3 - c^3\} + bc\{|d-a|^3 - a^3 - d^3\} \\ & + ad\{|b-c|^3 - b^3 - c^3\} + ac\{(b+d)^3 - b^3 - d^3\}. \end{aligned}$$

It vanishes when any argument is zero, and its mixed derivative is

$$\partial_a \partial_b \partial_c \partial_d N = 12\{\min(a,d) + \min(b,c)\} > 0.$$

The continuous second derivative of $|x|^3$ justifies this identity also across the two absolute-value crossings. Four integrations from zero prove $N > 0$ throughout the positive orthant.

For a nonzero centered law U, let μ_- and μ_+ be its negative-magnitude and positive restrictions to $(0, \infty)$ and put $h = \mathbb{E}U_+ = \mathbb{E}U_- > 0$. The measure

$$M(da, db) = \frac{a+b}{h}\, \mu_-(da)\mu_+(db)$$

has mass $\mathbb{P}(U \neq 0)$. At label (a, b) choose the centered two-point law on $\{-a, b\}$; with remaining mass $\mathbb{P}(U = 0)$ choose the zero law. Integrating its probabilities $b/(a+b), a/(a+b)$ recovers exactly

Law(U) and its absolute third moment. Use independent labels for U, W. The conditional difference is strictly positive at every pair of nonzero labels and zero if either label is zero. The nonzero-label pair has positive probability when both variables are nondegenerate. Integrating proves strictness; all moments are finite because $|u+w|^3 \le 4(|u|^3+|w|^3)$. The zero case gives equality. □

Corollary 3.4. *Every nonconstant original at a full global maximum over admissible representations is indecomposable as a convolution of two nonconstant independent real laws. The same statement holds at a full fixed-horizon global maximum if a deterministic-zero original slot is available. A genuine three-point original with raw arithmetic-progression support $\{0, g, 2g\}$ and positive weights (a, b, c) is therefore excluded in these classes when $b^2 \ge 4ac$.*

Proof. If a centered original X factors in law as $U_0 + W_0$, its finite third moment gives finite third moments of both factors. For example choose B with $\mathbb{P}(|W_0| \le B) > 0$ and use independence to bound

$$\mathbb{P}(|W_0| \le B)\mathbb{E}(|U_0| - B)_+^3 \le \mathbb{E}|U_0 + W_0|^3.$$

Center each factor separately. Their sum still has law X, their variances add to $\mathrm{Var}(X)$, and Lemma 3.3 strictly lowers the original price. Replacing X by these two independent originals leaves the full sum law, variance and positive selected discrepancy fixed and strictly improves the ratio. Reindexing the countable class permits the additional coordinate; a fixed finite horizon requires the stated zero slot. The Lévy price is additive under residual convolution, so this argument concerns originals.

For completeness, a genuine three-point real law admits such a factorization exactly when its support is an arithmetic progression and $b^2 \ge 4ac$. Independent factor supports A, B obey $A + B \subset \mathrm{supp}(X)$. Fixing a point of either support makes both finite, and $|A + B| \ge |A| + |B| - 1$ forces both to have two points. Their positive gaps must coincide: otherwise their four pair sums are distinct. For raw support $\{0, g, 2g\}$ put $m = b + 2c$. The Bernoulli probabilities are the roots of $t^2 - mt + c = 0$, with discriminant $b^2 - 4ac$. If this is nonnegative, both roots are in $(0, 1)$: their sum and product are positive, the polynomial at 1 is $a > 0$, and their sum is less than 2. Their independent sum has exactly the specified weights. Conversely two equal-gap bits give those weights and discriminant $(p-q)^2 \ge 0$. Thus splitting is admissible in exactly the stated arithmetic case. Nonarithmetic ternaries are indecomposable by the support argument; their optimality is not established here. □

With no free zero slot, a fixed-horizon ceiling does not license the additional original.

3.3 Atoms of a countable original sum

Lemma 3.5. *If the full sum has an atom of mass $m > 0$ and each X_i has at most three support points, then, with $q_i = \max_u \mathbb{P}(X_i = u)$,*

$$\sum_i (1 - q_i) < \infty.$$

Proof. Represent each X_i as an activated fair two-point law. If $q_i \ge 1/2$, activate with probability $\rho_i = 2(1 - q_i)$ and use the modal state with the remaining probability. On activation choose fairly between the modal state and an independent conditional nonmodal state. If three masses p_1, p_2, p_3 are all below $1/2$, mix the three fair pairs, assigning weight $1 - 2p_k$ to the pair that excludes state k. Thus in all cases $\rho_i = \min\{1, 2(1 - q_i)\} \ge \frac{3}{2}(1 - q_i)$.

Condition on activations and chosen endpoints, orienting the fair bits with positive increments. Every fixed-sum fiber is an antichain. Sperner's theorem [17] bounds its conditional probability

by $2^{-N}\binom{N}{\lfloor N/2\rfloor} \le \sqrt{2/(N+1)}$ when N bits are active. For a finite prefix with $\mu = \sum_i \rho_i > 0$, Cauchy–Schwarz and the generating function of N yield

$$\sup_x \mathbb{P}(\textstyle\sum_i X_i = x)^2 \le 2\mathbb{E}\frac{1}{N+1} = 2\int_0^1 \prod_i (1-\rho_i+\rho_i t)\,dt \le \frac{2(1-e^{-\mu})}{\mu}.$$

Convolution cannot increase the largest atom. The left side is at least m^2 for every prefix, so μ stays bounded, proving the assertion. This is a direct finite-alphabet concentration argument within the classical theory of infinite convolutions [11]. □

4 Eliminating jumps and nonshrinking supports

4.1 Pair contacts for the Lévy measure

Lemma 4.1. *For nonzero same-sign points $x, y \in \operatorname{supp}\nu$, one has $Q(x+y) = 0$. The measure ν is supported on at most two positive and two negative points.*

Proof. Choose a probability law H dominated by a finite multiple of ν and supported in a compact interval on one side of zero. Write $m = \mathbb{E}H$, $v = \mathbb{E}H^2$, $r = \mathbb{E}|H|^3$, and $k = m^2/v$. Remove a compensated Poisson component of intensity $p - kp^2$ with jump law H, and insert the single centered variable $B_pH - pm$. This is admissible for all sufficiently small $p > 0$ and preserves variance. Its third absolute moment is

$$pr - 3p^2|m|v + 4p^3|m|^3 - 2p^4|m|^3.$$

At the centered threshold $z - kp^2m$, the raw CDF is

$$[(I + pU_H)e^{-(p-kp^2)U_H}F](z), \qquad U_Hf(w) = \mathbb{E}f(w-H) - f(w).$$

The bounded-operator expansion is valid in uniform norm. Its second-order term is $(kU_H - \frac{1}{2}U_H^2)F$. Combining it with the normal and third-moment terms, the coefficient of p^2 in the excess is

$$(k+1)\mathbb{E}Q(H) - \tfrac{1}{2}\mathbb{E}Q(H+H'),$$

where H' is independent. The cubic cancellation uses $\mathbb{E}|H+H'|^3 = 2r + 6|m|v$. The first expectation vanishes by Lemma 3.1; the second is nonpositive. Maximality forces it to be zero. Upper semicontinuity extends this identity to all same-sign support pairs, by choosing H on neighborhoods of any two specified support points. The case $x = y$ is included.

A pair sum obtained here is a zero of Q; it need not belong to the Lévy support. The propagation argument below retains an actual old two-sided direction and inserts a new original at such a fixed contact.

For the support count, perturb the Lévy measure by a finite signed measure $\eta = f\nu$ with bounded f supported away from zero. Define $d_\eta = \int y\,d\eta$, $v_\eta = \int y^2\,d\eta$, and $r_\eta = \int |y|^3\,d\eta$. The centered threshold $z - d_\eta$ gives

$$(1+v_\eta)^{3/2}\left\{[e^{U_\eta}F](z) - \Phi\left(\frac{z-d_\eta}{\sqrt{1+v_\eta}}\right)\right\} - C(\beta + r_\eta) \le 0, \tag{4.1}$$

where $U_\eta f(w) = \int [f(w-y) - f(w)]\,\eta(dy)$. Both signs are allowed near zero. Negative perturbations are interpreted by removing the corresponding finite Poisson component; the bounded operator identity remains exact. Thus its Hessian is negative semidefinite on these signed directions.

Write $U_x f(w) = f(w-x) - f(w)$. Let $L = (15D + \phi'''(z))/4$, and, for $\sigma \in \{-1, 1\}$, define

$$\mathcal{K}_\sigma(u,v) = uv\left\{-3D + \left(3C + \frac{\sigma\phi''(z)}{2}\right)(u+v) - Luv + \frac{3C}{2}uv(u+v)\right\}, \quad u, v > 0. \tag{4.2}$$

At same-sign contact points $x = \sigma u$, $y = \sigma v$ whose sum is also a contact, the intensity Hessian equals $\mathcal{K}_\sigma(u,v)$. For verification before substituting contacts, its kernel is

$$\begin{gathered} U_xU_yF(z) + \tfrac{3}{2}[x^2U_yF(z) + y^2U_xF(z)] + B_0x^2y^2 \\ + E_0(x^2y + xy^2) + z\phi xy. \end{gathered}$$

Substitute $U_xF(z) = -\phi x - Ax^2 + C|x|^3$ and $U_xU_yF(z) = -2Axy + 3Cxy(|x| + |y|)$ to obtain (4.2). Its moment coefficient matrix in the basis (u, u^2, u^3) is

$$M_\sigma = \begin{pmatrix} -3D & 3C + \sigma\phi''(z)/2 & 0 \\ 3C + \sigma\phi''(z)/2 & -L & 3C/2 \\ 0 & 3C/2 & 0 \end{pmatrix}. \tag{4.3}$$

It is not negative semidefinite: its value at $(0, t, 1)$ is $-Lt^2 + 3Ct > 0$ for small $t > 0$.

If three distinct support magnitudes u_1, u_2, u_3 existed on one side, restrict ν to three disjoint small compact neighborhoods and normalize each restriction to a probability measure. Arbitrary signed combinations are locally admissible intensity directions. Their Hessian is congruent to M_σ through the columns of first, second and third moments of these restrictions. As the neighborhoods shrink, these columns tend to (u_j, u_j^2, u_j^3), an invertible Vandermonde matrix. For sufficiently small neighborhoods the Hessian cannot be negative semidefinite. This contradiction proves the support count. Finite second moment then makes ν a finite measure. □

4.2 Contacts from limiting support values

Lemma 4.2. *Suppose i_n are distinct and $u_n \in \operatorname{supp} X_{i_n}$ satisfies $u_n \to x \neq 0$. Then $Q(x) = Q(2x) = 0$. The probability transfer can be taken from a state $v_n \to 0$ to u_n, with gap $d_n = u_n - v_n \to x$. If the full CDF has a jump and $d_n > x$ along a subsequence, it has no jump at $z - x$ or $z - 2x$.*

Proof. The variances of distinct coordinates tend to zero. If F is continuous, choose a support state v_n with $|v_n| \le \sqrt{v_{i_n}}$. Deleting one or two selected coordinates changes the sum by a variable tending to zero in L^2. The deleted CDFs therefore converge uniformly to F. The derivatives in (3.4) satisfy

$$d_n \to x, \quad b_n = u_n^2 - v_n^2 \to x^2, \quad r_n' \to |x|^3, \quad r_n'' \to -6|x|^3.$$

First stationarity gives $Q(x) = 0$. In (3.5), the limit of the off-diagonal entry minus that of the diagonal entry is

$$F(z-2x) - 2F(z-x) + F(z) + 2Ax^2 - 6C|x|^3 = Q(2x).$$

The two-coordinate contrast $(1, -1)$ gives $-2Q(2x) \le 0$. Together with $Q \le 0$, this proves the assertion.

If the full law has an atom, Lemma 3.5 gives modal probabilities $q_n \to 1$. Take v_n to be a modal state; then $q_n v_n^2 \le v_{i_n}$, so $v_n \to 0$. Put $H_n(w) = \mathbb{P}(S - X_{i_n} + v_n \le w)$. Since the original law contains this branch with probability q_n,

$$\|H_n - F\|_\infty \le 1 - q_n.$$

The analogous double-deletion bound is $(1-q_n)+(1-q_m)$. Pass to a subsequence on which the gaps equal x, lie above x, or lie below x. Each shifted CDF then has a definite side limit. Let Q_1^*, Q_2^* denote the corresponding versions of $Q(x), Q(2x)$, using these side limits. Stationarity gives $Q_1^* = 0$; the same Hessian contrast gives $Q_2^* \geq 0$, while $Q \leq 0$ gives $Q_2^* \leq 0$. Closed CDF values dominate their left limits, so $Q(x), Q(2x) \geq 0$. Both are zero. When the arguments approach from the left, the two side limits must equal the closed values; this proves the last assertion. □

4.3 Comparisons with fixed added summands

Fix a sign σ and a nonzero limit $x = \sigma u$ as in Lemma 4.2. Its probability transfers are two-sided at each finite index. Appending a fresh raw variable $(1-t)\delta_0 + t\delta_y$, $t \geq 0$, has first derivative $Q(y)$ in (3.1). The gap y below is always fixed, rather than set equal to a varying support value. Thus when $Q(y) = 0$, its first derivative is exactly zero for every selected index.

Lemma 4.3. *For $y = \sigma v$ with $Q(y) = 0$, the limiting mixed Hessian between the old transfer tending to x and a fresh variable at y equals*

$$\mathcal{K}_\sigma(u, v) + Q(x + y).$$

The old diagonal and its mixed entry with a fresh variable at x both equal $\mathcal{K}_\sigma(u, u)$ in the limit. The same conclusions hold if the old direction instead varies the positive intensity of an existing Lévy atom at x.

Proof. The limiting derivatives of an old probability transfer are

$$m' = x, \quad B' = x^2, \quad B'' = -2x^2, \quad R_*' = |x|^3, \quad R_*'' = -6|x|^3.$$

They coincide with the derivatives for the fresh variable at x. Stationarity and (3.5) give the common limiting diagonal $\mathcal{K}_\sigma(u, u)$. Their mixed CDF double shift is closed: in the continuous case this follows from uniform deletion convergence; in the atomic case its argument approaches $z - 2x$ from the right, or Lemma 4.2 removes the left-limit jump. Since $Q(x) = Q(2x) = 0$, its mixed entry is the same diagonal.

Append also a fresh variable at the fixed contact y. Pass to a subsequence of the bounded Hessian entries. Write B_{xy} for the old–fresh-y entry and N_{xy} for the fresh-x–fresh-y entry. The latter uses the exact closed value $F(z - x - y)$ and, by the same substitution as in (4.2), equals $\mathcal{K}_\sigma(u, v) + Q(x + y)$. The old single-shift limit is fixed by stationarity; the fresh single shift is closed. Upper semicontinuity of the double shift, together with the uniform deletion or modal-branch comparison in the preceding lemma, therefore gives $N_{xy} - B_{xy} \geq 0$.

The vector $w = (-1, 1, 0)$ has zero limiting quadratic value and is feasible: it decreases an old probability and increases its fresh copy. For every $t \geq 0$, $w + t(0, 0, 1)$ is also a feasible tangent direction. The Hessian inequality gives

$$2t(N_{xy} - B_{xy}) + t^2 H_{yy} \leq 0.$$

Letting $t \downarrow 0$ proves the reverse inequality. This identifies the closed mixed entry. All first derivatives were exactly zero before taking the limit, so no approximation of a first-order contact is used to infer a second-order inequality.

For an existing Lévy atom, vary its intensity in both directions. Its CDF and moment derivatives are already at fixed closed shifts. Relative to the fresh Bernoulli diagonal its second derivative differs by

$$U_x^2 F(z) + 2Ax^2 - 6C|x|^3 = Q(2x) - 2Q(x) = 0,$$

using Lemma 4.1. This proves the same formulas. □

Proposition 4.4. *Every maximizing representation with three-point independent variables has* $\nu = 0$ *and* $\max\{|u| : u \in \operatorname{supp} X_i\} \to 0$.

Proof. Let $x = \sigma u$ be either a nonzero limiting support value or a point of $\operatorname{supp}\nu$. In the latter case it is an atom of positive intensity by Lemma 4.1. Retain this actual old two-sided probability or intensity direction throughout the argument. Let $y = \sigma v$ be any fixed same-sign contact of Q; it need not carry Lévy intensity or occur in an existing original. For fixed $\varepsilon > 0$, append *one* new raw three-point law

$$(1 - (1+\varepsilon)t)\delta_0 + t\delta_x + \varepsilon t\delta_y, \qquad t \geq 0.$$

Its first derivative is $Q(x) + \varepsilon Q(y) = 0$. If $m_1 = x + \varepsilon y$, $v_1 = x^2 + \varepsilon y^2$, and $s_1 = x|x| + \varepsilon y|y|$, its moment derivatives are

$$m' = m_1, \quad B' = v_1, \quad B'' = -2m_1^2, \quad R_*' = |x|^3 + \varepsilon|y|^3, \quad R_*'' = -6m_1 s_1.$$

Both nonzero states belong to this same new original. Its raw CDF is affine in t, so there is no raw double-shift term in its diagonal. Together with the displayed exact recentering derivatives, this gives the Hessian diagonal

$$\mathcal{K}_\sigma(u,u) + 2\varepsilon\mathcal{K}_\sigma(u,v) + \varepsilon^2\mathcal{K}_\sigma(v,v).$$

By Lemma 4.3, its mixed entry with the old direction is $\mathcal{K}_\sigma(u,u) + \varepsilon[\mathcal{K}_\sigma(u,v) + Q(x+y)]$. Evaluate the two-parameter limiting Hessian on $(-1,1)$, decreasing the old direction and increasing the new activation. It gives

$$-2\varepsilon Q(x+y) + \varepsilon^2\mathcal{K}_\sigma(v,v) \leq 0.$$

First take the old-coordinate limit with ε fixed, and only then let $\varepsilon \downarrow 0$. Thus $Q(x+y) \geq 0$, and hence $Q(x+y) = 0$. The admissible neighborhoods of the old probabilities need not have a uniform radius.

Starting with $Q(x) = 0$, this conclusion propagates to $Q(nx) = 0$ for every positive integer n, always using the same old direction and a fixed contact at the next step. At step $y = nx$, that contact is used only in the newly inserted ternary; no mass or intensity at nx is removed. Thus the induction requires no residual support membership for its pair-created contacts. This contradicts the bounded contact set in Lemma 3.1. No such x exists. The finite Lévy measure must therefore be zero. The uniformly bounded independent supports cannot have a nonzero accumulation along distinct indices, so their radii tend to zero. □

5 Excluding a Gaussian component

By Proposition 4.4, it is enough to consider

$$S = W + G_a, \qquad W = \sum_i X_i, \qquad \operatorname{Var}(W) = 1 - a, \quad \beta = \sum_i \mathbb{E}|X_i|^3,$$

where $G_a \sim N(0,a)$ is independent. Suppose $a > 0$. The law has a smooth density f. Threshold and Gaussian-variance stationarity give

$$f(z) = \phi(z), \qquad f'(z) = -3D - z\phi(z), \qquad |f''(z)| \leq 6C. \tag{5.1}$$

The second identity follows from the heat equation and the derivatives of (3.1). Both signs of a small variance change are allowed. The last inequality follows by expanding Q at zero on its two sides.

5.1 A smoothing bound from zero bias

For a centered variable T of variance $v > 0$, its zero-biased law T^* is characterized by $\mathbb{E}[Tg(T)] = v\mathbb{E}g'(T^*)$. Goldstein and Reinert [8] introduced this transformation and the independent-sum replacement construction. Goldstein [7, Lemma 1.1] and Tyurin [18, Theorem 3 and Corollary 1] give

$$W_1(W, W^*) \le \frac{\beta}{2(1-a)}, \tag{5.2}$$

where W_1 is Wasserstein distance with cost $|x-y|$. For countably many summands the statements below follow by finite approximation retaining the fixed Gaussian variance.

Lemma 5.1. *For every such independent sum,*

$$\Delta(W + G_a) \le K(a)\beta, \qquad K(a) = \frac{\phi(0)}{2}\int_0^1 \frac{s^2}{[1-(1-a)s^2]^{3/2}}\,ds.$$

The function K is decreasing, and $K(1/10) < C_{\mathrm{E}}$. Hence a maximizing representation has $a < 1/10$.

Proof. For $h(u) = \Phi((z-u)/\sqrt{a})$ and $v = 1-a$, the solution of $vg'(u) - ug(u) = h(u) - \mathbb{E}h(\sqrt{v}Z)$ has the Ornstein–Uhlenbeck representation

$$g(u) = -\int_0^1 \mathbb{E}h'(su + \sqrt{v(1-s^2)}Z)\,ds.$$

Differentiating twice and convolving Gaussian derivatives gives $\|g''\|_\infty \le \phi(0)\int_0^1 s^2(1-vs^2)^{-3/2}\,ds$. The zero-bias identity and (5.2) bound the expectation difference by $\beta\|g''\|_\infty/2$. The integral decreases with a, and direct integration gives

$$K(a) = \frac{\phi(0)}{2}\left\{\frac{1}{(1-a)\sqrt{a}} - \frac{\arcsin\sqrt{1-a}}{(1-a)^{3/2}}\right\}.$$

At $a = 1/10$ this is $(5\sqrt{10}/27)\phi(0)(3 - \arctan 3)$. To check the strict comparison without decimal rounding, use $\pi > 157/50$, $\sqrt{10} < 19/6$ and

$$\arctan 3 = \pi/2 - \arctan(1/3) > \frac{157}{100} - \left(\frac{1}{3} - \frac{1}{81} + \frac{1}{1215}\right) > \frac{237}{190} > \frac{21}{10} - \frac{27}{10\sqrt{10}}.$$

This is equivalent to $K(1/10) < C_{\mathrm{E}}$. □

5.2 A lattice comparison

Lemma 5.2. *At a maximizing representation with $a > 0$,*

$$\phi(z) \le a\left(C - \frac{f''(z)^2}{36C}\right). \tag{5.3}$$

Proof. We first approximate $W + G_a$ by a lattice convolution plus a Gaussian, with error smaller than its lattice spacing. Order the X_i by decreasing variance and set

$$v^{(K)} = \sum_{i>K} v_i, \qquad H_K = \sum_{i>K} v_i^2.$$

Then $KH_K \le v^{(K)} \to 0$. If the series is infinite, choose $h_K = \sqrt{H_K/K}$, so $H_K = o(h_K)$ and $Kh_K \to 0$. For finitely many nonzero summands take $h_K = K^{-2}$ beyond them.

Replace the tail after K by its accompanying compensated Poisson law with measure $\mu_K = \sum_{i>K} \mathrm{Law}(X_i)|_{\mathbb{R}\setminus\{0\}}$. Variance and additive third moment are unchanged. By (2.1), after convolution with G_a the uniform CDF error is at most

$$\frac{H_K}{8\pi}\int_0^\infty t^3 e^{-at^2/2}\,dt = \frac{H_K}{4\pi a^2} = o(h_K).$$

Extra powers of t give the corresponding density-derivative estimates.

Round each of the first K summands to its neighboring points on $h_K\mathbb{Z}$, independently and with its conditional mean unchanged. Writing the rounding error as δ, we have $\mathbb{E}(\delta \mid X) = 0$, $|\delta| \le h_K$ and $\mathbb{E}(\delta^2 \mid X) \le h_K^2/4$. Subtract its total added variance from the Gaussian component. Gaussian Taylor estimates bound the CDF change by $O_a(Kh_K^2) = o(h_K)$ and likewise control fixed density derivatives. The third-moment change is at most

$$3h_K^2 \sum_{i\le K} \mathbb{E}|X_i| + Kh_K^3 = o(h_K),$$

since the sum of first absolute moments is at most $\sqrt{K}$.

Replace the jumps of μ_K smaller than h_K in magnitude by their Gaussian variance, and round all larger jumps without changing their conditional means. Subtract their added variance from the Gaussian part. For a large jump y, with rounding variance $e_h(y)$, the compensated exponent error is bounded by

$$\left|\mathbb{E}e^{it(y+\delta)} - e^{ity} + \tfrac{1}{2}t^2 e_h(y)\right| \le |t|^3\left(\tfrac{1}{2}h^2|y| + \tfrac{1}{6}h^3\right).$$

For a small jump it is at most $|t|^3|y|^3/6$. Integration against μ_K gives exponent error $O(|t|^3 h_K v^{(K)})$ and third-moment change $O(h_K v^{(K)}) = o(h_K)$. The Gaussian variance changes by $o(1)$ and remains at least $a/2$. Factoring a common smaller Gaussian factor, the bound $|e^u - e^v| \le |u - v|$ for $\Re u, \Re v \le 0$ justifies the CDF and derivative comparisons by Fourier inversion.

We have obtained variance-one representations $T_j + G_{a_j}$, where T_j lies on a translate of $h_j\mathbb{Z}$, $a_j \to a$, and

$$\beta_j = \beta + o(h_j), \quad \|F_{T_j+G_{a_j}} - F\|_\infty = o(h_j), \quad f_j^{(k)} \to f^{(k)} \text{ uniformly}, \quad k = 0,1,2. \tag{5.4}$$

The rounded jump measure has finite intensity, because the pre-rounded jumps have magnitude at least h_j and finite second integral. Its centering is well-defined. All approximations are admissible representations.

Fix $p \in (0,1)$. Choose integers n_j and $p_j \to p$ so that $n_j h_j^2 p_j(1-p_j) = a_j$; rounding $a_j/[h_j^2 p(1-p)]$ to a nearby integer and solving for p_j does this, with the upper integer used when $p = 1/2$. Put $u_j = 1-2p_j$ and $B_j = h_j(\mathrm{Bin}(n_j,p_j) - n_j p_j)$. The lattice Edgeworth expansion at closed endpoints is

$$\mathbb{P}(B_j \le x) = \Phi_{a_j}(x) + \frac{h_j}{2}\phi_{a_j}(x) - \frac{a_j h_j u_j}{6}\phi''_{a_j}(x) + o(h_j), \tag{5.5}$$

uniformly in those endpoints. Here Φ_b, ϕ_b denote the centered normal CDF and density of variance b. This is the binomial case of the classical lattice expansion of Esseen; the closed-endpoint convention is recorded, for example, in Dinev and Mattner [5, equation (3)]. For the uniformity with $p_j \to p$, Stirling's formula gives, with $\sigma_j = \sqrt{n_j p_j(1-p_j)}$ and $t_k = (k - n_j p_j)/\sigma_j$,

$$\mathbb{P}(\mathrm{Bin}(n_j,p_j) = k) = \frac{\phi(t_k)}{\sigma_j}\left[1 + \frac{u_j(t_k^3 - 3t_k)}{6\sigma_j} + O\left(\frac{1+|t_k|^6}{n_j}\right)\right]$$

uniformly for $|t_k| \le \sqrt{B \log n_j}$ and p_j in a fixed compact subinterval. Chernoff bounds control the omitted tails by $o(n_j^{-1/2})$ when B is sufficiently large. Summation of the remainder is $O((\log n_j)^3/n_j)$, and right-endpoint Euler–Maclaurin summation adds $\phi(t_k)/(2\sigma_j)$. Integration of the cubic Hermite term is $(1-t_k^2)\phi(t_k)$, proving (5.5).

Choose z_j on the full lattice of $T_j + B_j$ with $|z_j - z| \le h_j/2$. Conditional on every value of T_j, the endpoint $z_j - T_j$ is a closed binomial endpoint, so (5.5) gives

$$F_{T_j+B_j}(z_j) = F_{T_j+G_{a_j}}(z_j) + \frac{h_j}{2} f_j(z_j) - \frac{a_j h_j u_j}{6} f_j''(z_j) + o(h_j).$$

Threshold stationarity and (5.4) make its base excess $D + o(h_j)$. The actual third-moment sum of the binomial carrier is $a_j h_j (1 + u_j^2)/2$. Maximality therefore implies, for $u = 1 - 2p$,

$$\phi(z) - \frac{au}{3} f''(z) - Ca(1 + u^2) \le 0.$$

The inequality extends to $u \in [-1, 1]$ by continuity. Optimize at $u = -f''(z)/(6C)$, which lies in this interval by (5.1), to obtain (5.3). □

Gaussian convolution gives the semiconvexity inequality $f'(z)^2 \le f(z)f''(z) + f(z)^2/a$ by Cauchy–Schwarz on its Gaussian mixture formula. Set $c = \phi(z)/(aC)$. Lemma 5.2 gives $0 < c \le 1$ and $|f''(z)| \le 6C\sqrt{1-c}$. Hence

$$(3D + z\phi(z))^2 \le aC^2\{c^2 + 6c\sqrt{1-c}\} \le \frac{45}{16} aC^2.$$

The last maximum is at $c = 3/4$. Moreover, $|z|\phi(z) \le \sqrt{2\phi(0)/e}\sqrt{\phi(z)}$. Since $C > 2/5$, $\phi(0) < 2/5$ and $e > 27/10$, it follows that

$$\beta \le \left(\frac{\sqrt{5}}{4} + \frac{\sqrt{2\phi(0)/e}}{3\sqrt{C}} \right) \sqrt{a} < \frac{17}{20}\sqrt{a}.$$

For the final rational comparison one may use $\sqrt{5}/4 < 14/25$ and $\sqrt{20/27}/3 < 29/100$. Together with Lemma 5.1, this confines a possible Gaussian component to

$$0 < \beta < \frac{269}{1000}, \qquad \frac{400}{289}\beta^2 < a < \frac{1}{10}. \tag{5.6}$$

5.3 Fourier exclusion of the remaining window

Korolev and Shevtsova [12, Section 3.3, Theorem 2] prove that a centered compound-Poisson variable of variance $v > 0$ and third absolute jump integral r satisfies

$$d_K(Y, N(0, v)) \le 0.3051\, r/v^{3/2}. \tag{5.7}$$

Their jump law is assumed to have positive variance. A vanishing symmetric perturbation of a constant jump law removes this restriction. The same bound holds for every finite-variance infinitely divisible law with third jump integral r: truncate its Lévy measure and replace the omitted variance and Gaussian part by symmetric jumps of size η. These add at most $v\eta$ to the third jump integral. Pass to the limit using weak continuity of distance to the continuous normal CDF.

Lemma 5.3. *A variance-one independent sum satisfying (5.6) has $\Delta(S) < (2/5)\beta$.*

Proof. The elementary inequality

$$\cos x \le 1 - x^2/2 + |x|^3/10 \tag{5.8}$$

holds on the real line. For $0 \le x \le 12/5$ use the fourth-degree cosine upper Taylor bound, and for $x \ge 5$ its right side is at least one. On $12/5 \le x \le 5$, the inequality $\cos x \le -1 + (x-\pi)^2/2$ reduces the claim to positivity of $P(x) = 2 - x^2 + \pi x - \pi^2/2 + x^3/10$. With $u = x - 4$,

$$P(x) = P(4) + (\pi - 16/5)u + u^2(1/5 + u/10).$$

For $-8/5 \le u \le 0$ this is at least $P(4)$, and for $0 \le u \le 1$ it is at least $P(4) - (5/4)(\pi - 16/5)^2$. Both are positive from $157/50 < \pi < 22/7$. Evenness proves (5.8).

For independent centered copies X, X', the pointwise inequality

$$|x-y|^3 \le 2(|x|^3 + |y|^3) - 2xy(|x| + |y|)$$

gives $\mathbb{E}|X - X'|^3 \le 4\mathbb{E}|X|^3$. Thus a coordinate characteristic function h_i obeys

$$|h_i(t)| \le \exp\{-v_i t^2/2 + b_i|t|^3/5\}, \qquad b_i = \mathbb{E}|X_i|^3. \tag{5.9}$$

Write h for the full characteristic function. On omitting coordinate i, (5.9) and the exact Gaussian factor give

$$e^{-at^2/2} \prod_{j \ne i} |h_j(t)| \le e^{25/54 - t^2/2 + \beta t^3/5}, \qquad t \ge 0.$$

Indeed $b_i \ge v_i^{3/2}$ and $\max_{u \ge 0}(u^2/2 - u^3/5) = 25/54$. The single-coordinate form of (5.2) gives $|h_i(t) - h_i^*(t)| \le b_i t/(2v_i)$. Selecting the replaced coordinate with its variance probability in the zero-bias identity yields

$$|h'(t) + th(t)| \le \frac{\beta}{2} t^2 e^{25/54 - t^2/2 + \beta t^3/5}.$$

Multiply by $e^{t^2/2}$ and integrate from zero. Then

$$|h(t) - e^{-t^2/2}| \le \frac{5}{6} e^{25/54 - t^2/2}(e^{\beta t^3/5} - 1) \le \frac{\beta}{6} e^{25/54} t^3 e^{-t^2/2 + \beta t^3/5}. \tag{5.10}$$

These calculations first apply to a finite sum. Finite approximations retaining a and converging in variance and third-moment sum justify both inequalities for the countable sum.

Set $c = 19/10$ and $T = c/\beta > 7$. On $[0, 6]$ use $\beta \le 269/1000$, and on $[6, T]$ use $\beta t \le c$. It follows that

$$\frac{1}{\pi\beta} \int_0^T \frac{|h(t) - e^{-t^2/2}|}{t}\, dt \le \frac{e^{25/54}}{6\pi}(I + J), \tag{5.11}$$

where

$$I = \int_0^6 t^2 e^{-t^2/2 + (269/5000)t^3}\, dt, \qquad J = \int_6^\infty t^2 e^{-3t^2/25}\, dt.$$

For the high frequencies, $|h_W(t)|$ is bounded by the nonnegative characteristic function of a symmetric infinitely divisible variable Π with Lévy measure $\frac{1}{2}\sum_i \mathrm{Law}(X_i - X_i')$ off zero. This follows from $|h_i| \le \exp[-(1 - |h_i|^2)/2]$. Its variance is $V = 1 - a$ and its third jump integral is at most 2β. By (5.7), $\Delta(\Pi, N(0, V)) \le 0.6102\beta/V^{3/2}$. Integration by parts against a Gaussian density of variance $a/2$ gives

$$\begin{aligned} f_{\Pi + G_{a/2}}(0) &\le \frac{\phi(0)}{\sqrt{V + a/2}} + \frac{0.6102\beta}{V^{3/2}} \frac{\sqrt{2/\pi}}{\sqrt{a/2}} \\ &< \frac{\phi(0)}{\sqrt{0.95}} + \frac{0.6102}{0.9^{3/2}} \frac{17}{20} \frac{2}{\sqrt{\pi}} < \frac{11}{10}. \end{aligned}$$

The first inequality uses the total variation of the Gaussian density; the second uses (5.6). The nonnegative characteristic function of Π and exact density inversion now give

$$\frac{1}{\pi}\int_T^\infty \frac{e^{-at^2/2}|h_W(t)|}{t}\,dt \le \frac{e^{-aT^2/4}}{T} f_{\Pi+G_{a/2}}(0).$$

Consequently its ratio to β is less than $(11/19)e^{-361/289} < 1/6$. The corresponding normal tail is at most

$$\frac{1}{\pi\beta}\int_T^\infty \frac{e^{-t^2/2}}{t}\,dt \le \frac{\beta}{\pi c^2} e^{-c^2/(2\beta^2)} < 10^{-9}.$$

Appendix A supplies the complete rational checks

$$I < 23/10, \quad J < 3/8, \quad e^{25/54}/(6\pi) < 17/200,$$

and the two scalar high-frequency comparisons just used. Exact CDF inversion is applicable because the sum has Gaussian variance $a > 0$. Combining its two frequency ranges proves

$$\frac{\Delta(S)}{\beta} < \frac{17}{200}\left(\frac{23}{10}+\frac{3}{8}\right)+\frac{1}{6}+10^{-9} = \frac{1182125003}{3000000000} < \frac{2}{5}.$$

□

Equations (5.6) and Lemma 5.3 contradict $C > C_{\mathrm{E}} > 2/5$. This proves $a = 0$ in Theorem 1.2.

6 An infinite maximizing sum is nonatomic

We now have $S = \sum_i X_i$, with uniformly bounded supports whose radii tend to zero. Suppose this full law has an atom. We prove that only finitely many variables can be nonconstant.

Choose a modal state c_i of probability q_i, and set $p_i = 1 - q_i$. Lemma 3.5 gives $\sum_i p_i < \infty$. Centering gives

$$|c_i| \le \sqrt{v_i p_i/q_i}, \qquad \sum_i |c_i| \le \sqrt{\sum_i v_i}\sqrt{\sum_i p_i/q_i} < \infty,$$

since $q_i \ge 1/3$. Write $X_i = c_i + Y_i$. With probability one only finitely many Y_i are nonzero, and every finite configuration of positive-probability states is an atom of the full law, because $\prod_i q_i > 0$.

Two compactness observations retain the endpoint conventions needed below. For fixed M, the set of sums using at most M nonzero Y_i is compact. In a sequence of such sums, each of its at most M selected indices can be made either constant or divergent; divergent indices contribute zero in the limit. Every limit is another finite configuration. If all the selected nonzero values are positive, then, for each $b > 0$, these sums avoid an interval $[b-\eta, b)$ for some $\eta > 0$. Otherwise a sequence approaching b from below has fixed terms whose sum is b and escaping positive terms, so it is eventually at least b, a contradiction.

If the maximizing threshold z is itself an atom of mass $m > 0$, all sufficiently late gaps from the mode to another state are negative. Indeed, for a positive gap $d_i = u_i - c_i \to 0$, the modal-branch CDF $H_i(w) = \mathbb{P}(S - X_i + c_i \le w)$ satisfies $\|H_i - F\|_\infty \le p_i \to 0$. Hence

$$H_i(z - d_i) - H_i(z) \longrightarrow -m.$$

The stationarity equation (3.4) instead makes this difference tend to zero, since both states and their gap tend to zero. This excludes such a sequence.

Fix K so that $q_j \geq 3/4$ for $j \geq K$, and select a nonmodal state of an arbitrarily late nonconstant coordinate $i \geq K$. Set $d_i = u_i - c_i$ and $S_i^0 = S - X_i + c_i$. Its exact mass-transfer derivative is

$$g_i = -\operatorname{sgn}(d_i)\mathbb{P}(E_i), \qquad E_i = \begin{cases} \{z - d_i < S_i^0 \leq z\}, & d_i > 0, \\ \{z < S_i^0 \leq z - d_i\}, & d_i < 0. \end{cases} \tag{6.1}$$

Let N_i count the nonmodal outcomes among $j \geq K$, $j \neq i$. For every fixed integer M,

$$E_i \subset \{N_i > M\} \quad \text{for all sufficiently late } i. \tag{6.2}$$

If z is not an atom, it has positive distance from the compact set of all configurations with at most $M + K$ nonmodal states. The intervals in (6.1) shrink to z, proving (6.2). If z is an atom, take K beyond the positive-gap exclusion. Conditional on any of the finitely many early prefix values, write $S_i^0 = b - \sum_{j \geq K, j \neq i} Z_j$ with $Z_j \geq 0$. Its pivotal interval is exactly

$$\sum_{j \geq K, j \neq i} Z_j \in [b - z - |d_i|, b - z).$$

For $b - z \leq 0$ it is empty; otherwise the one-sided compactness observation proves (6.2). The right endpoint is excluded, as required.

Vary every nonmodal probability for $j \geq K$, $j \neq i$, by the factor $1 + t$, taking the compensating mass from the mode. Both signs of $|t| < 1/2$ are admissible. The original product-law score is

$$L_i = \sum_{j \geq K, j \neq i} \ell_j, \qquad \ell_j = \begin{cases} 1, & Y_j \neq 0, \\ -p_j/q_j, & Y_j = 0. \end{cases} \qquad L_i \geq N_i - \Lambda, \quad \Lambda = \sum_{j \geq K} p_j/q_j < \infty.$$

This infinite product variation is twice differentiable for every bounded observable. Its density is

$$(1 + t)^{N_i} \prod_{Y_j = 0} (1 - t p_j/q_j).$$

The product and its first two derivatives are dominated, uniformly in i, by a constant times a polynomial in N_i times $(3/2)^{N_i}$. These bounds are integrable because $\mathbb{E} b^{N_i} \leq \exp((b-1)\sum_{j \geq K} p_j)$ for every $b \geq 1$. Differentiating (6.1) in this direction gives the exact mixed raw-CDF derivative

$$g_{i,t} = -\operatorname{sgn}(d_i)\mathbb{E}[\mathbf{1}_{E_i} L_i].$$

By (6.2), $\mathbb{E}[L_i \mid E_i] \to \infty$ whenever these events have positive probability.

Let $h_i = \max\{|x| : x \in \operatorname{supp} X_i\} \to 0$. The mass-transfer stationarity equation and

$$|u_i^2 - c_i^2| \leq 2h_i|d_i|, \qquad |r_i'| \leq 6h_i^2|d_i|$$

give

$$g_i/d_i \to -\phi(z), \qquad \mathbb{P}(E_i)/|d_i| \to \phi(z) > 0. \tag{6.3}$$

It follows that $|g_{i,t}|/|d_i| \to \infty$.

On the other hand, form (3.1) with two parameters: the selected mass transfer s in coordinate i and the collective variation t just defined. Uniformly in late i, its Hessian satisfies

$$|J_{ss}| \leq K_1 d_i^2, \qquad |J_{st} - g_{i,t}| \leq K_2|d_i|, \qquad |J_{tt}| \leq K_3. \tag{6.4}$$

To check these bounds, the selected derivatives are $m_s = d_i$, $B_s = u_i^2 - c_i^2$, $B_{ss} = -2d_i^2$, $F_{ss} = 0$, $F_s = O(|d_i|)$ and $|R_{ss}| \le 18h_i d_i^2$ by (3.4). All collective moment derivatives through order two are bounded by summability of p_j and the common support bound. The raw CDF derivatives are uniformly bounded by the product estimate above. Mixed moment derivatives vanish because the two parameters affect disjoint coordinates. The normal normalization is smooth near $(B, m) = (1, 0)$, proving (6.4). Both parameters are two-sided, so negative semidefiniteness gives $|J_{st}|^2 \le K_1 K_3 d_i^2$. This contradicts the divergent mixed CDF ratio. Therefore an infinite maximizing sum is nonatomic.

Conversely, if $\sum_i (1 - q_i) < \infty$, the same modal-drift argument and Borel–Cantelli show that the series is a countable discrete law. This proves the modal-deficit assertion and completes the proof of Theorem 1.2.

7 Support motions and the completion rank

We prove Theorem 1.5 directly at the maximizing representation. Suppose, to the contrary, that finite sets of arbitrarily late coordinates admit nonconstant state functions with constant sum on their completion sets. Subtract their individual means, absorbing the sum into the constant c. Write

$$\mathbb{E} f_i(X_i) = 0, \quad w_i = \mathbb{E} f_i(X_i)^2, \quad W = \sum_{i \in I} w_i > 0, \quad V = \sum_{i \in I} v_i, \quad h = \max_{i \in I} \max |\operatorname{supp} X_i|.$$

Along the supposed sequence, $h + V \to 0$.

Move each raw state x_i to $x_i + t f_i(x_i)$ and move the raw threshold to $z + tc$. Every completion pattern has the same residual threshold as before. For each other pattern the residual threshold lies outside the closed support of T_I, in an open interval where its CDF is constant. There are finitely many patterns, so one two-sided interval of sufficiently small t works for all of them. Distinct coordinate states remain distinct after reducing the interval if necessary. Thus the raw CDF is exactly unchanged, also when a completion is an atom or the endpoint of a gap. This reasoning permits noncompact remainder support.

Choose j maximizing w_j / v_j among the nonconstant functions, and put

$$X = X_j, \quad f = f_j(X_j), \quad v = v_j, \quad w = w_j,$$
$$m = \mathbb{E} X f, \quad a_1 = \mathbb{E} X^2 f, \quad b_1 = \mathbb{E} X f^2, \quad M = \sum_{i \in I} \mathbb{E}[X_i f_i(X_i)].$$

Then

$$v / w \le V / W. \tag{7.1}$$

Tilt the original law of coordinate j to $(1 + sf)\, d\operatorname{Law}(X)$, and apply the support motion. Both signs of s are admissible because its support is finite. Its moved mean is $sm + stw$. The centered threshold x and total variance B have derivatives

$$\begin{aligned} x_s = -m, \quad x_t = c, \quad x_{st} = -w, \qquad B_s = a_1, \quad B_{ss} = -2m^2, \\ B_t = 2M, \quad B_{tt} = 2W, \qquad B_{st} = 2b_1. \end{aligned} \tag{7.2}$$

The exact raw CDF satisfies $F_{ss} = F_t = F_{st} = F_{tt} = 0$. Cauchy–Schwarz gives

$$|m| \le \sqrt{vw}, \quad |a_1| \le h\sqrt{vw}, \quad |b_1| \le hw, \quad |M| \le \sqrt{VW}.$$

For clarity, the separately recentered third moments have the following derivatives; all expectations involving X, f concern the selected coordinate:

$$\begin{aligned} R_t &= 3\sum_{i\in I}\mathbb{E}[X_i|X_i|f_i], & R_{tt} &= 6\sum_{i\in I}\mathbb{E}[|X_i|f_i^2],\\ R_s &= \mathbb{E}[|X|^3 f] - 3m\mathbb{E}[X|X|], & R_{ss} &= 6m^2\mathbb{E}|X| - 6m\mathbb{E}[X|X|f],\\ R_{st} &= 3\mathbb{E}[X|X|f^2] - 3w\mathbb{E}[X|X|] - 6m\mathbb{E}[|X|f]. \end{aligned} \tag{7.3}$$

It follows that

$$\begin{aligned} &|R_t| \le 3h\sqrt{VW}, && |R_{tt}| \le 6hW,\\ &|R_s| \le 4h^2\sqrt{vw}, && |R_{ss}| \le 12hvw, && |R_{st}| \le 12h^2 w. \end{aligned}$$

These bounds use $\mathbb{E}X^4 \le h^2 v$ and do not require a lower bound on any state probability.

First stationarity in (3.1) gives

$$\phi c = 2AM - CR_t, \qquad F_s = -\phi m - Aa_1 + CR_s.$$

Thus $|c| \le K\sqrt{VW}$ and $|F_s| \le K\sqrt{vw}$. Using (7.2)–(7.3) in its Hessian now gives

$$|J_{ss}| \le K_1 vw, \qquad |J_{tt}| \le K_2 W, \qquad J_{st} = \phi w + O((h+V)w).$$

Indeed the term $-\phi x_{st}$ is ϕw, direct variance and third-moment terms are $O(hw)$, and products of first derivatives are $O(\sqrt{VWvw}) = O(Vw)$ by (7.1). All constants depend only on the fixed maximizing representation and threshold. Negative semidefiniteness would imply, for sufficiently late sets,

$$\phi^2 w^2/4 \le K_1 K_2 vwW \le K_1 K_2 V w^2,$$

contradicting $V \to 0$. This proves Theorem 1.5.

Proof of Corollary 1.6. Fix a late cut and absorb all earlier blocks into the finite prefix; let it have N distinct sum values. Select $M > 2N$ consecutive remaining blocks. The unchanged infinite future has support in an interval of positive width $W = \sum_{l \text{ after these blocks}} H_l$. Distinct vectors of selected block-sum values have total sums at least W apart: at their first differing block k, their difference in absolute value is at least

$$\delta_k - \sum_{l>k,\ l \text{ selected}} H_l \ge W.$$

For each prefix atom, at most two such vectors put the maximizing threshold inside the translated closed future interval. Thus at most $2N$ selected block vectors can complete through the actual remainder support, including endpoints.

There are $M+1$ unknowns $a_1, \dots, a_M, c$ and at most $2N$ homogeneous equations $\sum_k a_k u_k = c$ on these vectors. A solution exists with some $a_k \neq 0$. Apply Theorem 1.5 to the individual functions $f_i(x) = a_k x$ for coordinates in block k. At least one is nonconstant because each block is nonconstant. Their sum is constant on every completion pattern, a contradiction for a sufficiently late cut. The total variance and largest coordinate radius of the selected finite family tend to zero, independent of M. □

8 Support shape and weighted variance

We now impose the additional third-moment bound

$$\beta \le \beta_* := \frac{18481}{25000}. \tag{8.1}$$

It supplies the strict scalar curvature inequalities proved below. Throughout these sections we use a maximizing representation with at most three support points per original coordinate, as in Theorem 1.2, with data (1.5). Omit deterministic coordinates and put

$$H_i = \operatorname{diam}(\operatorname{supp} X_i), \qquad v_i = \mathbb{E}X_i^2, \qquad V(h) = \sum_{H_i \le h} v_i, \qquad W_s = \sum_i v_i H_i^{-s}.$$

Constants denoted by K may depend on this fixed representation and on explicitly fixed exponents, but not on a selected set of its coordinates or on a scale tending to zero.

Lemma 8.1 (The scalar curvature margin)**.** *Let $C \ge C_{\mathrm{E}}$, $D = C\beta$, and define*

$$L_0 = \frac{15D + \phi'''(z)}{4}, \qquad c_\varepsilon = 3C + \varepsilon\phi''(z)/2, \quad \varepsilon \in \{-1, 1\}.$$

Under (8.1), $c_\varepsilon > 0$ and $c_\varepsilon^2 > 3DL_0$ for both signs.

Proof. Set $r = 486833/500000$. The identity $\phi(0)/C_{\mathrm{E}} = 6/(3+\sqrt{10}) < r$ follows by squaring the positive number $6/r - 3$. With $x = |z|$, a lower bound for $(c_\varepsilon^2 - 3DL_0)/C^2$ is

$$\left[3 - \frac{r}{2}|x^2 - 1|e^{-x^2/2}\right]^2 - \frac{3\beta_* r}{4}|3x - x^3|e^{-x^2/2} - \frac{45}{4}\beta_*^2.$$

All square brackets are positive. On $0 \le x \le 1/2$, replace $e^{-x^2/2}$ by its upper bound $T(x) = 1 - x^2/2 + x^4/8 - x^6/48 + x^8/384$. This gives the polynomial

$$P(x) = \left[3 - \tfrac{r}{2}(1 - x^2)T(x)\right]^2 - \tfrac{3\beta_* r}{4}(3x - x^3)T(x) - \tfrac{45}{4}\beta_*^2.$$

Here is a finite rational positivity check, specified entirely by these formulas. Write $P(x) = \sum_{j=0}^{20} p_j x^j$ and start with $b_i = \sum_{j \le i} p_j 2^{-j}\binom{i}{j}/\binom{20}{j}$. Successive midpoint subdivision replaces a coefficient row by the left and right edges of its averaging triangle. On the five indicated intervals the smallest resulting coefficient exceeds the stated rational number:

I	$[0,1/8]$	$[1/8,3/16]$	$[3/16,7/32]$	$[7/32,1/4]$	$[1/4,1/2]$
$\min b_i(I) >$	$1/40$	$1/1000$	$1/30000$	$1/1000$	$9/1000.$

The corresponding subdivision words are $00, 010, 0110, 0111, 1$. The Bernstein basis is nonnegative and sums to one, so these finite rational comparisons prove $P > 0$ on $[0, 1/2]$. For $x \ge 1/2$, elementary differentiation gives $|\phi''(x)|/\phi(0) < 2/3$ and $|\phi'''(x)|/\phi(0) < 7/5$. The remaining lower bound is

$$(3 - r/3)^2 - \tfrac{3\beta_* r}{4}\tfrac{7}{5} - \tfrac{45}{4}\beta_*^2 = \frac{35773617637}{140625000000} > 0.$$

Finally $3C - \phi(0)/2 > 0$ proves positivity of both c_ε. □

Lemma 8.2 (Distances of the extreme support points from zero)**.** *There is $\kappa > 0$ such that every sufficiently late genuine three-point coordinate, with extreme states $-a_i, b_i$, satisfies*

$$\min(a_i, b_i) \geq \kappa H_i.$$

Consequently, after enlarging a fixed constant to include the finite prefix,

$$H_i \mathbb{E}|X_i| \leq K v_i. \tag{8.2}$$

For a two-point coordinate the latter identity holds with $K = 2$.

Proof. For a genuine triple $x < y < t$ in $[-h, h]$, vary its raw mean and second moment, keeping its support fixed. The probability map is nonsingular because its Vandermonde determinant is nonzero. If r is its centered third absolute moment and $k = [u|u|]_{x,y,t}$, divided differences give

$$|k| \leq 1, \quad |r_\mu| \leq 12h^2, \quad |r_v| \leq 3h, \quad |r_{\mu\mu}| \leq 30h, \quad r_{\mu v} = -3k, \quad r_{vv} = 0.$$

For example the last two identities follow by differentiating $\mathbb{E}|Y - \mu|^3$ and using the mass, first-moment and second-moment coordinates; the second divided difference of $u|u|$ lies between -1 and 1. The exact objective second variation, obtained as in (3.1), is

$$(-3D - Cr_{\mu\mu})e^2 + (\phi''(z) + 6Ck + 3Cr_\mu)ef + (-L_0 + 3Cr_v)f^2.$$

It is nonpositive for all real e, f. Since $h \to 0$, any limit of k satisfies $(3Ck + \phi''(z)/2)^2 \leq 3DL_0$. By Lemma 8.1, k stays a fixed distance from both -1 and 1 on late triples.

Write their extremes as $-a, b$. For $y \geq 0$ and $y \leq 0$ respectively,

$$k = 1 - \frac{2a^2}{(a+y)(a+b)}, \qquad k = -1 + \frac{2b^2}{(a+b)(b-y)}.$$

These formulas and centering give the asserted bound on both extremes. If the middle point is nonnegative, the negative extreme alone carries $\mathbb{E}|X|/2$ in first moment; hence $v \geq a\mathbb{E}|X|/2 \geq \kappa H \mathbb{E}|X|/2$. Use the positive extreme when the middle point is nonpositive. For $X = H(B_p - p)$, $\mathbb{E}|X| = 2p(1-p)H$ and $v = p(1-p)H^2$. □

9 Third-moment growth under grouping and normal comparison

A group is a sum of disjoint original coordinates. Its cost is the increase in the sum of their third absolute moments. This cost is essential even though grouping does not change the full sum law.

Lemma 9.1 (Two grouping estimates)**.** *For a centered independent family with total variance V,*

$$0 \leq \mathbb{E}|\textstyle\sum X_i|^3 - \sum \mathbb{E}|X_i|^3 \leq 2V^{3/2}. \tag{9.1}$$

For independent raw variables Y_i, let $X_i = Y_i - \mathbb{E}Y_i$, $M = \sum \mathbb{E}|Y_i|$, and $A_- = \sum \mathbb{E}(Y_i)_-$. Then their grouping cost is at most $2M^3 + 6A_-V$. If the coordinates are divided into B classes, with centered first-moment and variance sums M_j, V_j, there is, for every integer $J \geq 1$, a partition into at most $8J + 6B$ groups whose cost is at most

$$\frac{18M^3}{J^2} + \frac{9S^2}{J}, \qquad M = \sum_j M_j, \quad S = \sum_j \sqrt{M_j V_j}. \tag{9.2}$$

All bounds pass to countable groups with finite summed variance and third-moment sum.

The first upper bound is a classical third-moment estimate. Pinelis [14, equation (5)] gives the sharper coefficient $\mathbb{E}|Z|^3 = 2\sqrt{2/\pi}$ in place of 2. The elementary proof below suffices for all the rates used here.

Proof. The function $\mathbb{E}|x+Y|^3 - |x|^3$ is convex for centered Y, because its second derivative is $6(\mathbb{E}|x+Y| - |x|) \geq 0$. Jensen proves the lower bound. The pointwise Taylor bound

$$|x+y|^3 \leq |x|^3 + 3x|x|y + 3|x|y^2 + |y|^3$$

gives an upper increment $3v_i\sqrt{\sum_{j<i} v_j}$ at step i. Summing and comparing with $\int_0^V 3\sqrt{u}\,du$ proves (9.1). Applied to tail differences, the same inequality proves convergence in L^3.

For centered X define $k_X(t) = \mathbb{E}(X-t)_+ - (-t)_+$. It is nonnegative, has supremum $\mathbb{E}|X|/2$, and integral $v_X/2$. Taylor's integral remainder yields the exact identity

$$\mathbb{E}|X+Y|^3 - \mathbb{E}|X|^3 - \mathbb{E}|Y|^3 = 12\int_{\mathbb{R}} k_X(t)k_Y(-t)\,dt.$$

For a raw variable with first-moment data M_i, A_i, $k_{X_i}(t) \leq M_i$ and $k_{X_i}(-t) \leq (M_i - t)_+ + A_i$ for $t \geq 0$. Split the integral at zero and apply the same bounds to each preceding partial sum. The increment is at most

$$6(M_{<i}M_i^2 + M_iM_{<i}^2) + 6(A_iV_{<i} + A_{<i}v_i).$$

The first terms telescope to at most $2M^3$ and the others sum to at most $6A_-V$.

More generally choose raw laws and signs with $X_i = \varepsilon_i(Y_i - \mathbb{E}Y_i)$ and keep the two signs in separate groups. Retain individually every coordinate with $M_i > M/J$ or $A_i > A_-/J$. Greedily accumulate the remaining weights $M_i/M + A_i/A_-$ until they reach $1/J$, omitting the second weight when $A_- = 0$. Every completed group has both normalized weights at most $3/J$. There are at most $4J+2$ groups, including the retained coordinates and the final group in each sign. The total cost is at most $18M^3/J^2 + 18A_-V/J$.

In class j take $Y_i = X_i$, so $A_- = M_j/2$, and allocate $J_j = \lceil J(M_j/M + \sqrt{M_jV_j}/S)\rceil$. Then $\sum J_j \leq 2J + B$ and the two terms sum to (9.2). Upper budgets in place of M, S only decrease the sum of the allocated weights. These observations also justify removal of any selected original coordinates before grouping. □

Lemma 9.2 (A quadratic majorant for a sum of coordinates)**.** *Let U be any finite or countable sum of original coordinates, independent of a remainder with CDF G, and let $\mathbb{E}U^2 = \sigma^2 > 0$. There is a quadratic polynomial $P(u) = a + bu + cu^2$ such that*

$$Q_U(u) := P(u) + C|u|^3 - G(z-u) \geq 0, \qquad \mathbb{E}Q_U(U) \leq 2C\sigma^3. \tag{9.3}$$

If $\sigma \to 0$, the original coordinate radii are at most H, and $H/\sigma \to 0$, its coefficients can be chosen with

$$|a| + |b| + |c| \leq K, \qquad b + \phi(z) = O(\sigma), \qquad c + A = O(\sigma). \tag{9.4}$$

In particular $Q_U(u) \leq K(1 + |u|^3)$ globally.

Proof. Put $f(u) = G(z-u) - C|u|^3$ and maximize $\mathbb{E}f(Y)$ at prescribed first and second moments (m, s) with $s > m^2$. The value is a finite concave function: $f \leq 1$, and a two-point law gives a finite lower bound. A supporting plane at $(0, \sigma^2)$ gives a quadratic majorant of f at every real point. To include point masses without a continuity assumption, mix δ_x with a law approaching the value at $(0, \sigma^2)$, apply the plane inequality at the interior moment pair, and cancel its mixing weight. This

yields $P(x) \geq f(x)$. Replacing the original block by any such Y is an admissible one-coordinate comparison. If its third-moment sum is β_I, maximality bounds the value by $F(z) - C\beta_I$, whereas the actual grouped law has value $F(z) - C\mathbb{E}|U|^3$. Their difference is at most $C(\mathbb{E}|U|^3 - \beta_I) \leq 2C\sigma^3$, proving (9.3).

We give the coefficient argument because it fixes the quadratic term used later. Deleting the original block and applying the universal inequality to its remainder gives, for $\sigma^2 \leq 1/2$,

$$G(z-u) \leq \Phi((z-u)/\sqrt{1-\sigma^2}) + C(\beta - \beta_I)(1-\sigma^2)^{-3/2}.$$

Normal Taylor bounds imply $G(z-u) - F(z) + \phi(z)u \leq K(u^2+\sigma^2)$. Its expectation is zero, so

$$\mathbb{E}|G(z-U) - F(z) + \phi(z)U| \leq K\sigma^2. \tag{9.5}$$

For any bounded score ψ with $|\psi| \leq 1$, $\mathbb{E}\psi(U) = \mathbb{E}U\psi(U) = 0$, replace the whole block by $(1+t\psi)\,\mathrm{Law}(U)$, $|t| \leq 1$. Write $g = \mathbb{E}G(z-U)\psi$, $d = \mathbb{E}U^2\psi$, $e = \mathbb{E}|U|^3\psi$. Its total variance and CDF are $1 + td, F(z) + tg$, and its third-moment sum is $\beta + w + te$, where $0 \leq w \leq 2\sigma^3$. Here $|g| \leq K\sigma^2$, $|d| \leq \sigma^2$. Taylor expansion of $B^{3/2}[f - \Phi(z/\sqrt{B})]$ at $(1, F(z))$, whose B derivative is A, has remainder $O(\sigma^4)$ for $|t| \leq 1$. Using both signs of t gives

$$|\mathbb{E}\{G(z-U) + AU^2 - C|U|^3\}\psi| \leq Cw + K\sigma^4.$$

The L^1 quotient duality for the subspace spanned by $1, U$ therefore supplies a_0, b_0 with

$$\mathbb{E}|G(z-U) - C|U|^3 - a_0 - b_0U + AU^2| = O(\sigma^3).$$

We may take $a_0 = F(z) + A\sigma^2 - C\mathbb{E}|U|^3$ by changing the error constant. Equation (9.5) and $\mathbb{E}|U| \geq c\sigma$ give $b_0 + \phi(z) = O(\sigma)$. The latter lower bound follows from convergence of U/σ to normal; the classical Berry–Esseen bound is at most KH/σ.

Subtract this polynomial from the majorant in (9.3). Its L^1 norm under U is $O(\sigma^3)$. On polynomials of degree at most two, the normal L^1 norm restricted to $[-2,2]$ controls all three coefficients. The same bound holds uniformly under U/σ once its Kolmogorov distance to normal is small: absolute quadratics on the unit coefficient sphere have uniformly bounded variation there. It follows that $a - a_0 = O(\sigma^3)$, $b - b_0 = O(\sigma^2)$, $c + A = O(\sigma)$, as required. □

Lemma 9.3 (Weighted normal comparison)**.** *In Lemma 9.2, suppose* $(H/\sigma)\log^{3/2}(1/\sigma) \to 0$. *Then*

$$\mathbb{E}Q_U(\sigma Z) \leq K(\sigma^3 + H), \qquad Z \sim N(0,1). \tag{9.6}$$

If $H/\sigma = O(\sigma^\zeta)$ *for a fixed* $\zeta > 1$, *then for every* $2 < q < \min(\zeta+1, 3)$ *and fixed* T,

$$\sup_{|t| \leq T\sigma} |F(z+t) - F(z) - \phi(z)t + At^2| = O_{q,T}(\sigma^q). \tag{9.7}$$

Proof. We record the weighted approximation estimate used in the proof. For a variance-one sum W of centered independent variables bounded by ϵ, and $\epsilon(1+L^3)$ sufficiently small,

$$|\mathbb{P}(W \leq x) - \Phi(x)| + |\mathbb{P}(W < x) - \Phi(x)| \leq K\epsilon(1+x^2)\phi(x), \qquad |x| \leq L. \tag{9.8}$$

Here is the bounded-array exponential-tilting proof. If $K(t) = \log \mathbb{E}e^{tW}$, centered summands under the tilted product law are bounded by 2ϵ, so $|K'''(t)| \leq 2\epsilon K''(t)$ and $e^{-2\epsilon|t|} \leq K''(t) \leq e^{2\epsilon|t|}$. For $x \geq 0$ choose $K'(t) = x$. Integration gives $t = x + O(\epsilon x^2)$, $s^2 = K''(t) = 1 + O(\epsilon x)$, and

$K(t)-tx=-x^2/2+O(\epsilon x^3)$. Berry–Esseen under the tilted law, applied to $e^{-tsu}\mathbf{1}_{\{u\ge 0\}}$ or its strict version, gives

$$\mathbb{P}(W\ge x)=e^{K(t)-tx}\{J(ts)+O(\epsilon)\},\quad J(a)=\int_0^\infty e^{-au}\phi(u)\,du.$$

The same formula holds with $>$. Since $J(a)\ge c/(1+a)$, $|(\log J)'(a)|\le\sqrt{2/\pi}$, and $e^{-x^2/2}J(x)=1-\Phi(x)$, the relative tail error is $O(\epsilon(1+x^3))$. Mills' bound and reflection prove (9.8). Finite truncations and thresholds tending to x from both sides prove the countable version. This is the classical Cramér tilting argument; no nonlattice assumption enters. For a modern independent-array formulation and its conjugate-law proof, see Beknazaryan, Sang and Xiao [1, Theorem 2.1]. Here the bounded-summand estimate above supplies the hypotheses directly.

Put $Q_\sigma(x)=Q_U(\sigma x)$, $w(x)=(1+x^2)\phi(x)$, and $L=\sqrt{20\log(1/\sigma)}$. In the sense of measures on $[-L,L]$,

$$dQ_\sigma=a'_\sigma(x)\,dx+d\mu(x),\quad d\mu\ge 0,\quad a_\sigma(x)=P(\sigma x)+C\sigma^3|x|^3.$$

The positive measure comes from $-G(z-\sigma x)$. Bounded coefficients give $\int w|a'_\sigma|\le K\sigma$. Since $Q_\sigma\ge 0$, integration by parts against w yields

$$\int_{-L}^{L} w\,|dQ_\sigma|\le K\{\sigma+(1+L^3)I+(1+L^2)\phi(L)\},\quad I=\int_{-L}^{L}Q_\sigma(x)\phi(x)\,dx.$$

Both endpoint terms are included. Stieltjes integration by parts and (9.8), with $\epsilon=H/\sigma$, now give

$$I\le \mathbb{E}Q_U(U)+K\epsilon\{\sigma+(1+L^3)I+(1+L^2)\phi(L)\}.$$

The term containing I is absorbed. The global cubic bound on Q_U pays the normal tails and proves (9.6). The left-continuous function $G(z-\sigma x)$ and its positive jump measure can be used throughout; the two endpoint versions in (9.8) justify the same argument when either law has atoms.

For a fixed $|t|\le T$, the normal density ratio on $[-L,L]$ under translation by t is at most $e^{TL+T^2/2}$. The preceding weighted-variation argument applied to $Q_\sigma(x-t)$, together with

$$\mathbb{P}(|U|>\sigma L)\le 2\exp\{-L^2/[2(1+\epsilon L/3)]\}=\sigma^{10-o(1)},$$

therefore gives

$$\sup_{|t|\le T}\mathbb{E}Q_U(U-\sigma t)\le\sigma^{-o(1)}(\sigma^3+H).\tag{9.9}$$

The omitted actual tail is bounded by Cauchy–Schwarz and $\mathbb{E}|U|^6=O(\sigma^6)$; all fixed normalized moments are bounded by the elementary bounded-summand exponential-moment estimate.

The coordinate contact identities determine the linear coefficient more accurately. Multiplying each identity by X_i and summing gives

$$\mathbb{E}[UG(z-U)]=-\phi(z)\sigma^2-A\mathbb{E}U^3+C\sum_i\eta_i,\quad \eta_i=\mathbb{E}[X_i|X_i|^3]-3v_i\mathbb{E}[X_i|X_i|].$$

Consequently

$$(b+\phi(z))\sigma^2=-(c+A)\mathbb{E}U^3-C\mathbb{E}[U|U|^3]+\mathbb{E}[UQ_U(U)]+C\sum_i\eta_i.$$

Here $|\mathbb{E}U^3|\le H\sigma^2$, $\sum|\eta_i|\le 4H^2\sigma^2$. Splitting at $|U|=\sigma\sqrt{M\log(1/\sigma)}$ and using nonnegativity, $\mathbb{E}Q_U(U)=O(\sigma^3)$ and bounded-summand tails gives $\mathbb{E}|U|Q_U(U)=O(\sigma^4\sqrt{\log(1/\sigma)})$. Thus $b+\phi(z)=O(\sigma^2\sqrt{\log(1/\sigma)})$ under the hypothesis $\zeta>1$. Finally the exact identity

$$F(z+\sigma t)-F(z)=-b\sigma t+c\sigma^2t^2+C\{\mathbb{E}|U-\sigma t|^3-\mathbb{E}|U|^3\}+\mathbb{E}Q_U(U)-\mathbb{E}Q_U(U-\sigma t)$$

and (9.9) prove (9.7). □

9.1 The closed-CDF phase identity

We use two elementary exponential bounds for independent centered variables. If their ranges have lengths d_i, integration of the tilted log-MGF bound $K_i''(t) \le d_i^2/4$ gives $\mathbb{E}e^{t\sum Y_i} \le \exp(t^2 \sum d_i^2/8)$, the bounded-range estimate of Hoeffding [10]. If instead $|Y_i| \le H$, the exponential series and $k! \ge 2 \cdot 3^{k-2}$ for $k \ge 2$ give

$$\log \mathbb{E}e^{t\sum Y_i} \le \frac{t^2 \sum \mathbb{E}Y_i^2}{2(1 - H|t|/3)}, \qquad H|t| < 3.$$

Chernoff's inequality gives the Bernstein bound used above and below. It also gives bounded moments of every fixed order after normalization when $H/\sqrt{\sum \mathbb{E}Y_i^2}$ is bounded.

Unbiased rounding to the adjacent points of a lattice of mesh ℓ preserves the mean and increases variance by at most $\ell^2/4$. For N independent coordinates, choose the first $N-1$ lattice phases uniformly and the last so that their sum is z modulo ℓ. For each fixed phase vector the rounded coordinates are independent. If their sum is $L_{\boldsymbol{u}}$ and their unrounded sum has CDF F, then

$$\mathbb{E}_{\boldsymbol{u}} F_{L_{\boldsymbol{u}}}(z) = \mathbb{E}F(z - W_\ell), \quad W_\ell = T_1 + \cdots + T_{N-1} - U_\ell, \tag{9.10}$$

where the T_i are independent with density $\ell^{-1}(1 - |x|/\ell)_+$ and U_ℓ is uniform on $[0, \ell]$. To verify the sign and endpoints, the phase-averaged mass at $z + k\ell$ is ℓ times the convolution of the N triangular kernels with the original law. For the last kernel, $\ell \sum_{k \le 0} k_\ell(z + k\ell - y)$ equals one for $y \le z$, $(z + \ell - y)/\ell$ for $z < y < z + \ell$, and zero for $y \ge z + \ell$. Tonelli proves (9.10), including atoms. In particular

$$\mathbb{E}W_\ell = -\ell/2, \qquad \mathbb{E}W_\ell^2 = (N+1)\ell^2/6.$$

For $N \ge 3$ every pair of phases is independent. If $v_{\boldsymbol{u}}$ is their total rounding variance, then

$$\mathbb{E}v_{\boldsymbol{u}} = N\ell^2/6, \quad \mathrm{Var}(v_{\boldsymbol{u}}) \le N\ell^4/64, \quad |\mathrm{Cov}(v_{\boldsymbol{u}}, F_{L_{\boldsymbol{u}}}(z))| \le \sqrt{N}\ell^2/16. \tag{9.11}$$

These follow by integrating $\ell^2 t(1-t)$ over a uniform fractional part and using $0 \le F_{L_{\boldsymbol{u}}} \le 1$.

Lemma 9.4 (Grouping at a selected normal window)**.** *Suppose* (9.7) *holds with* $2 < q < 3$ *at scales* $\sigma \downarrow 0$. *Suppose the original sum is partitioned into* $N \to \infty$ *independent groups with actual cost* $w \ge 0$. *The maximum is impossible if, with* $L = \log(1/\sigma)$,

$$N\sigma^{2q-2}L \to 0, \qquad w\sqrt{NL}/\sigma \to 0, \qquad wN^{2/3} \to 0. \tag{9.12}$$

Proof. The conditions permit a mesh

$$\max(w, \sigma^q) \ll \ell \ll \min(\sigma/\sqrt{NL}, N^{-2/3}).$$

Indeed $\sigma^q N^{2/3} \to 0$ follows from the first condition and $q < 4$. The triangular noise in (9.10) leaves $[-\sigma, \sigma]$ with probability at most $\exp\{-c\sigma^2/(N\ell^2)\}$; its first two tail moments satisfy the same bound up to polynomial factors. This is $o(\ell)$ because $\ell \ge \sigma^q$ and the exponent dominates L. Thus the window gives an average CDF increment $\phi(z)\ell/2 - A(N+1)\ell^2/6 + O(\sigma^q) + o(\ell)$.

For a fixed phase, the normalized objective expands as

$$(1+v)^{3/2}[F_{\boldsymbol{u}} - \Phi(z/\sqrt{1+v})] = F_{\boldsymbol{u}} - \Phi(z) + Av + \tfrac{3}{2}(F_{\boldsymbol{u}} - F(z))v + O(v^2).$$

Equations (9.11) cancel the term $-AN\ell^2/6$. The remaining covariance and normalization errors are $O(\ell^2 + \sqrt{N}\ell^2 + N\ell^3 + N^2\ell^4) + o(\ell)$. The actual cubic interpolation cost is at most $K\ell^2 \sum_i \mathbb{E}|Z_i| + KN\ell^3 \le K\sqrt{N}\ell^2 + KN\ell^3$, since the groups have total variance one. Every error and w is $o(\ell)$. The average objective is therefore positive, a contradiction. □

Lemma 9.5 (Small-diameter Fourier alternative)**.** *If $V(r) \ge cr^p$ for all sufficiently small r, for some fixed $p < 2$, the original representation cannot maximize.*

Proof. We give the approximation needed for this implication; smoothness of the original density alone would not suffice. Put $w(t) = \exp\{-t^2 V(1/(8t))/12\}$ for $t \ge 1$. The hypothesis makes $\int_1^\infty t^j w(t)\,dt$ finite for every fixed j. For a centered coordinate of diameter $d \le 1/(8t)$ and variance v, each of the following characteristic functions has modulus at most $e^{-vt^2/12}$, for $1 \le t \le \pi/\ell$: the original law, its accompanying compensated Poisson law, its unbiased rounding to $\ell\mathbb{Z}$, and the following rounding of its Poisson law. Interpolate the measure $x^2\,\mathrm{Law}(X)(dx)$ to $\ell\mathbb{Z}$; interpret its mass at zero as Gaussian variance and its mass at $u \ne 0$ as u^2 times a Poisson intensity.

For the original and Poisson versions use $1 - \cos s \ge s^2/4$ on $|s| \le 1$, with an independent copy for the original law. If $t\ell \le 1/4$, the directly rounded diameter is at most $d + 2\ell$, so the same estimate applies. Otherwise $d < \ell/2$, and the directly rounded law has masses $\rho/2, \rho/2$ at $-\ell, \ell$, with $\rho = \mathbb{E}|X|/\ell < 1/2$. Since $\mathbb{E}|X| \ge v/d \ge 8tv$ and $1 - \cos s \ge 2s^2/\pi^2$ on $[0, \pi]$, its damping is stronger than the stated bound. For the interpolated Poisson measure, $|tu| \le \pi + 1/8$ and $1 - \cos(tu) \ge t^2u^2/12$; the Gaussian mass at zero also has the required damping. Omitting one contributing factor loses at most $e^{1/768}$.

Order the original variances decreasingly, put $A_k = \sum_{i>k} v_i^2$, and choose $\ell_k = \sqrt{A_k/k}$. Then $A_k = o(\ell_k)$ and $k\ell_k \to 0$. Replace the tail by its accompanying Poisson law, interpolate its second-moment measure as above, and round the first k coordinates directly. The three full characteristic-function errors, with damping from all remaining factors, are bounded respectively by

$$KA_k t^4 w(t), \qquad K\ell_k^2 t^4 w(t), \qquad Kk\ell_k^2 t^2 w(t).$$

For the first use $|e^{f_i - 1} - f_i| \le v_i^2 t^4/8$. For the second use $(e^{itx} - 1 - itx)/x^2 = -t^2 \int_0^1 (1-s) e^{istx}\,ds$, whose second derivative is bounded by $t^4/12$. The interpolation preserves both total variance and the Poisson contribution to the additive third-moment quantity, because $|x|$ is affine between adjacent lattice nodes, including those adjoining zero. Direct rounding costs $O(k\ell_k^2)$ in variance and third-moment sum by bounded original supports.

The created Gaussian variance a_k is at most $\sum_{i>k} v_i \to 0$. Replace it by n fair centered Bernoullis of span ℓ_k, taking n nearest to $4a_k/\ell_k^2$. Their variance differs by $O(\ell_k^2)$ and their total third-moment sum is $O(a_k\ell_k) = o(\ell_k)$. For $|t\ell_k| \le \pi$, $|\cos(t\ell_k/2)|^n \le e^{\pi^2/16} e^{-a_k t^2/2}$; on $|t\ell_k| \le 1$, logarithmic expansion gives an additional full error $K\ell_k^2(t^2 + t^4)w(t)$. Outside that range use the damping itself. The final admissible law Z_k lies on a translate of $\ell_k\mathbb{Z}$, has variance $1 + o(\ell_k)$ and additive third-moment quantity $\beta + o(\ell_k)$, and satisfies

$$\int_0^{\pi/\ell_k} |\widehat{Z}_k(t) - \widehat{S}(t)|\,\frac{dt}{t} = o(\ell_k), \qquad \int_0^{\pi/\ell_k} |\widehat{Z}_k(t) - \widehat{S}(t)|\,dt = o(1).$$

The bounds below $t = 1$ use $w = 1$. All passages through infinite products are justified by the displayed summed variance bounds.

Fourier inversion gives a smooth density f for S. At a lattice point x, the exact mid-CDF formula is

$$F_{Z_k}(x) - \tfrac{1}{2}\mathbb{P}(Z_k = x) = \tfrac{1}{2} - \frac{\ell_k}{2\pi} \int_0^{\pi/\ell_k} \Im(e^{-itx}\widehat{Z}_k(t)) \cot(t\ell_k/2)\,dt.$$

The inequalities $0 \le (\ell/2)\cot(t\ell/2) \le 1/t$ and $|1/t - (\ell/2)\cot(t\ell/2)| \le K\ell^2 t$ yield $\mathbb{P}(Z_k = x) = \ell_k f(x) + o(\ell_k)$ and $F_{Z_k}(x) = F(x) + \ell_k f(x)/2 + o(\ell_k)$ uniformly at lattice points. At the maximizing threshold $f(z) = \phi(z) > 0$. Choose a lattice point within $\ell_k/2$ of z; its original discrepancy differs from D by $O(\ell_k^2)$. After the variance and third-moment corrections, its objective has gain $\phi(z)\ell_k/2 + o(\ell_k) > 0$. □

10 The first weighted variance moment

Lemma 10.1. *Under (8.2), $W_{1/2}$ and $W_{3/4}$ are finite.*

Proof. For a cutoff t set $M(t) = \sum_{H_i>t} \mathbb{E}|X_i|$ and

$$\mathcal{K}(t) = \sum_i v_i \min(1, t/H_i) = V(t) + t \sum_{H_i>t} v_i/H_i.$$

Then $M(t) \le K\mathcal{K}(t)/t$ and $V(t) \le \mathcal{K}(t)$. The exact Tonelli identity is

$$\sum_{H_i\le 1} v_i H_i^{-s} = V(1) + s\int_0^1 V(u)u^{-s-1}\,du. \tag{10.1}$$

We first prove $W_{1/2} < \infty$. If it diverges, good points with $V(u) \ge u^a$, $a = 13/24$, occur arbitrarily near zero. By Lemma 9.5 there are also $r \downarrow 0$ with $V(r) \le r^p$, $p = 15/8$. Take the largest good point below r, as a supremum. Monotonicity and right-continuity of the closed variance function give exactly

$$V(h) = h^a, \quad V(u) < u^a \ (h < u < r), \quad h \le r^{p/a}.$$

There is a gap below r because $V(r) < r^a$; approaching h from above and below proves equality even at a variance atom. Set $q = 23/8$, $t = h^{37/72}$. Then $h < t < r$, and integration by parts for the actual discrete variance measure gives

$$\mathcal{K}(t) \le t/r + t^a/(1-a).$$

Indeed the part with $H_i \le r$ equals $tV(r)/r + t\int_t^r V(u)u^{-2}\,du$. Select finitely many coordinates with $H_i \le h$ until their variance first reaches $h^{2/3}$. This is possible since $h^a > 2h^{2/3}$; the overshoot is at most h^2. Its actual standard deviation is $\sigma \sim h^{1/3}$ and Lemma 9.3 gives the window with exponent q. The two positive margins are

$$\sigma^{-2/3}\mathcal{K}(t) \le K(h^{1/360} + h^{97/1728}).$$

Thus, with $L = \log(1/\sigma)$, $(1 + M(t))\sigma^{q-2}L \to 0$ and $V(t)^3 L/\sigma^2 \to 0$.

To check the grouping consequence explicitly, take $J = \lceil (1 + M(t))\sigma^{-q} \rceil$ and retain the first $\lfloor \sqrt{J} \rfloor$ original coordinates separately. Group the remaining large coordinates by Lemma 9.1, and the small ones by variance: retain coordinates exceeding $V(t)/J$ and accumulate the others into groups of variance at most $2V(t)/J$. The complete count is $N = O(J)$, $N \to \infty$, and

$$w \le K\{(1 + M)^3/J^2 + (1 + M)/J + V^{3/2}/\sqrt{J}\}.$$

The three expressions in (9.12) tend to zero. For the second, the middle term is bounded by $K\sqrt{(1 + M)\sigma^{q-2}L}$, the first is smaller by $(1 + M)\sigma^q$, and the last is $KV^{3/2}\sqrt{L}/\sigma$. For the third, the middle term is $((1 + M)\sigma^{q-2}L)^{2/3}\sigma^{(4-q)/3}L^{-2/3}$, and the same two comparisons apply. This contradicts Lemma 9.4.

Now write $W = W_{1/2} < \infty$ and suppose $W_{3/4} = \infty$. Repeat the crossing argument with

$$a = 7/8, \quad p = 15/8, \quad Q_0 = 15/7, \quad q = 29/14, \quad t = \sqrt{h}.$$

It gives $V(h) = h^a$, $V(u) < u^a$ on (h, r) and $h \le r^{p/a}$. Select original variance $\sigma^2 \in [h^{2/Q_0}, h^{2/Q_0} + h^2]$. The window is available because $q < Q_0 < 3$. For the large-coordinate first moments and dyadic class sum,

$$M(t) \le K\{h^{-7/30} + h^{-1/16}\}, \qquad S(t) = \sum_j \sqrt{M_j V_j} \le KW, \qquad V(t) \le h^{7/16}.$$

The head estimate uses $M(H > r) \le KW/\sqrt{r}$ and the gap estimate uses $\int_t^r V(u)u^{-2}\,du$.

Here is the more precise class allocation needed at this step. For a window of exponent q, the annular cost $K(M^3/J^2 + S^2/J + V^{3/2}/\sqrt{J})$ is sufficient if

$$M\sigma^{q-4/3}L^{2/3} \to 0, \quad S^2\sigma^{q-2}L \to 0, \quad V^3L/\sigma^2 \to 0, \quad B\sigma^{2q-2}L \to 0. \tag{10.2}$$

In fact put $U_0 = \sigma^{2-2q}/L$, $D_0 = 1 + B + M^2L^{1/3}\sigma^{-2/3} + S^4L\sigma^{-2}$, and $J = \lceil\sqrt{D_0U_0}\rceil$. Then $D_0 \ll J \ll U_0$. Applying (9.2), the variance partition, and the same retained prefix verifies the first two costs in (9.12). The third follows by multiplying $w\sqrt{JL}/\sigma$ by $\sigma J^{1/6}/\sqrt{L} \le \sigma^{(4-q)/3}L^{-2/3}$. This proves the criterion.

For the displayed crossing parameters, $(q - 4/3)/Q_0 = 31/90 > 7/30, 1/16$, $(q - 2)/Q_0 = 1/30$, and $3a/2 - 2/Q_0 = 91/240$. Also $B = O(\log(1/t)) = O(L)$. All conditions in (10.2) hold with strict power margins, giving the second contradiction. □

11 A quantitative probability variation

Write the nonnegative full-law contact defect as

$$d(u) = -Q(u) = F(z) - \phi(z)u - Au^2 + C|u|^3 - F(z - u).$$

To control inverse convolution without a continuity hypothesis, fix $m > 3$ and use

$$E_m(h) = h^3 + \sup_{r \ge h}(h/r)^m \sup_{|u| \le r} d(u). \tag{11.1}$$

Then $h^3 \le E_m(h) \le K$ for $h \le 1$ and $E_m(th) \le t^m E_m(h)$ for $t \ge 1$. The added h^3 merely includes the cubic polynomial in the same bound.

Lemma 11.1 (Actual same-sign contacts)**.** *Translate a modal raw state of each of two distinct original coordinates to zero, writing its law as $K_i = (1 - \theta_i)\delta_0 + \eta_i$, $\theta_i \le 1/4$. Let the diameters be at most Lh, and choose actual states $x = \varepsilon u$, $y = \varepsilon v$, with $\kappa h \le u, v \le Lh$. Under Lemma 8.1, uniformly as $h \to 0$,*

$$d(x) \le K\theta_1 E_m(h), \quad d(y) \le K\theta_2 E_m(h), \quad d(x + y) \le K(\theta_1 + \theta_2)E_m(h).$$

The bounds do not involve the probabilities of the selected states.

Proof. Write $T_a f(u) = f(u + a)$ and $L_i = (T_{x_i} - I)K_i^{-1}$. The inverse is the absolutely convergent series $(1 - \theta_i)^{-1}\sum_{n\ge0}[-T_{\eta_i}/(1 - \theta_i)]^n$. A term translated by n raw states costs at most $K(n + 1)^m E_m(h)$ against d. Summing the geometric coefficients proves that removing one inverse changes a finite difference by $O(\theta_i E_m(h))$. Against $g(u) = |u|^3$ the same bound has h^3 in place of $E_m(h)$.

Let m_i be the raw mean and let r_i, s_i be the first and second derivatives of its centered cubic moment when mass moves from zero to x_i. Direct differentiation gives

$$\begin{aligned} v_i' &= x_i^2 - 2m_i x_i, \\ r_i &= |x_i - m_i|^3 - |m_i|^3 - 3x_i\mathbb{E}[(Y_i - m_i)|Y_i - m_i|], \\ s_i &= -6x_i\{(x_i - m_i)|x_i - m_i| + m_i|m_i|\} + 6x_i^2\mathbb{E}|Y_i - m_i|. \end{aligned}$$

The exact first score is $L_i f(0) = -\phi x_i - A v_i' + C r_i$, where $f(u) = F(z-u)$. Since $L_i u = x_i$, $L_i u^2 = v_i'$, this is $L_i d(0) = C(L_i g(0) - r_i) = O(\theta_i h^3)$. Removing the inverse proves the individual bounds.

Keep the raw event fixed by compensating the centered threshold by minus the changing mean. The original probability Hessian is negative semidefinite, however small the two feasible probability intervals. Divide its ij entry by $x_i x_j$, a congruence. With $a_i = v_i'/x_i$, $b_i = r_i/x_i$, its exact entries are

$$M_{ii} = -3D + \phi'' a_i - L_0 a_i^2 + 3C a_i b_i - C s_i/x_i^2,$$
$$M_{12} = -3D + \tfrac{\phi''}{2}(a_1 + a_2) - L_0 a_1 a_2 + \tfrac{3C}{2}(a_1 b_2 + a_2 b_1) + \{C L_1 L_2 g(0) - L_1 L_2 d(0)\}/(xy).$$

These follow by substitution in (3.5); only probabilities and normal normalization are differentiated. Uniformly, $a_i = x_i + O(\theta_i h)$, $b_i = \varepsilon |x_i|^2 + O(\theta_i h^2)$, and $s_i/x_i^2 = -6|x_i| + O(\theta_i h)$. The double cubic difference is $L_1 L_2 g(0) = 3uv(u+v) + O((\theta_1+\theta_2)uvh)$: its derivative under translation is bounded by $6|xy|$. The double defect difference is $d(x+y) - d(x) - d(y) + O((\theta_1+\theta_2)E_m(h))$.

Consequently the deterministic part is the symmetric kernel

$$B_\varepsilon(u,v) = -3D + c_\varepsilon(u+v) - L_0 uv + \tfrac{3C}{2} uv(u+v).$$

The diagonal errors are $O(\theta_i h)$ and the off-diagonal upper error is $K(\theta_1+\theta_2)E_m(h)/(uv)$; the omitted term $-d(x+y)/(uv)$ is nonpositive. Test on $(1+t,-1)$, $t = c_\varepsilon(u-v)/(3D)$. Its two entries have opposite signs, so that upper off-diagonal bound gives a lower bound on the quadratic form. Direct expansion gives

$$(1+t,-1)B_\varepsilon(1+t,-1)^{\mathsf{T}} = (u-v)^2\{c_\varepsilon^2/(3D) - L_0 + O(h)\}.$$

Every polynomial term retains $(u-v)^2$, including when the gaps coincide. The strict scalar margin gives $uv(u-v)^2 \le K(\theta_1+\theta_2)E_m(h)$. Now retain the defect and test on $(1,-1)$. The polynomial form is $(u-v)^2[-L_0 + 3C(u+v)]$, and the defect contributes $2d(x+y)/(uv)$. The preceding estimate absorbs the polynomial term and proves the pair bound. □

Lemma 11.2 (A selected-intensity bound)**.** *Fix $\kappa, L > 0$. Consider finitely many actual coordinates with modal raw states zero, diameters at most Lh, and modal deficits at most ϵ. In one physical sign, select submeasures of their actual nonmodal states in $[\kappa h, Lh]$, of total mass $0 < \lambda \le 1$. For sufficiently small h, ϵ,*

$$\epsilon E_m(h) \ge c\lambda h^4 \min(\lambda, h), \qquad \lambda \le K\{\sqrt{\epsilon E_m(h)}\, h^{-2} + \epsilon E_m(h) h^{-5}\}. \tag{11.2}$$

The independent remainder and all unselected states are retained.

Proof. Denote the selected submeasures by ν_i, their masses by r_i, and their normalized positive mark law by $\mu = \lambda^{-1}\sum \nu_i$. Write $a = \int x\, d\mu$, $b = \int x^2\, d\mu$, $c_3 = \int x^3\, d\mu$. They have their actual orders h, h^2, h^3 and $a \ge \kappa h$. For sign ε put $s = 2c_\varepsilon a/(3D) = O(h)$, $\alpha = 2 - s$. For $q > 0$ small use the count distribution with generating function

$$B_q(t) = e^{\alpha q(t-1)}[1 + sq(t-1) + q(t-1)^2]. \tag{11.3}$$

This is a probability law. To check positivity, its bracket is $1 + (1-s)q - \alpha q t + q t^2$. With $x = \alpha q$, its zeroth and first coefficients are positive, the latter being $e^{-x}\alpha q^2(1-s)$. For $n \ge 2$, after extracting $e^{-x}x^{n-2}q/(n-2)!$, the coefficient is $1 - \alpha^2 q/n + \alpha^2 q^2(1-s)/(n(n-1)) > 0$. Here $0 \le s \le 1$ and a fixed sufficiently small upper bound on q suffices.

Append one centered coordinate $\varepsilon(\sum_{j=1}^{N_q} Z_j - 2qa)$, with iid marks of law μ. In original coordinate i, withdraw fraction $\alpha q/\lambda$ of ν_i to its own modal state. All laws stay positive for $0 \le q \le q_0\lambda$, with a fixed $q_0 > 0$. Take the appended coordinate independently of the modified coordinates. For the raw convolution calculation below, use the carrier $\varepsilon \sum_{j=1}^{N_q} Z_j$ and keep the raw threshold fixed. The admissible coordinate is its recentered version $\varepsilon(\sum_{j=1}^{N_q} Z_j - 2qa)$.

We bound the full convolution error, since an uncontrolled cubic Taylor remainder would destroy the estimate. In the weighted convolution norm $\|\xi\|_m = \int (1 + |u|/h)^m \, |\xi|(du)$, put $D_\mu = T_{\varepsilon\mu} - I$ and $A_i = \lambda^{-1}(T_{\varepsilon\nu_i} - r_i I)K_i^{-1}$. The modal series gives

$$\sum_i \|A_i\|_m \le K, \quad \|D_\mu - \sum_i A_i\|_m \le K\epsilon, \quad \sum_i \|A_i\|_m^2 \le K\epsilon/\lambda.$$

The relative convolution operator is

$$\mathcal{P}(q) = \mathcal{R}(q)[I + q(sD_\mu + D_\mu^2)], \quad \mathcal{R}(q) = \prod_i (I - \alpha q A_i)e^{\alpha q D_\mu}.$$

All factors commute. Its logarithmic derivative is $\alpha(D_\mu - \sum A_i) - \alpha^2 q \sum A_i^2 (I - \alpha q A_i)^{-1}$. Thus $\|\mathcal{R}'(0)\|_m \le K\epsilon$ and $\sup_{q \le q_0\lambda} \|\mathcal{R}''(q)\|_m \le K\epsilon/\lambda$, whence

$$\|\mathcal{P}(q) - I - q\mathcal{P}'(0)\|_m \le K\epsilon q^2/\lambda. \tag{11.4}$$

Against d this costs $K\epsilon q^2 E_m(h)/\lambda$.

The moments of the fresh law are also exact. Its first three factorial count moments are $2q$, $2q + (4 - s^2)q^2$, and $6\alpha q^2 + (8 - 6s^2 + 2s^3)q^3$. Let $P_0 = \mathbb{P}(N_q = 0)$. Since every positive marked value is at least $\kappa h > 2qa$, expansion on the zero and positive events gives

$$\begin{aligned} \mathrm{Var}(\textstyle\sum Z_j) &= 2q(b + a^2) - s^2 a^2 q^2, \\ \mathbb{E}|\textstyle\sum Z_j - 2qa|^3 &= q(2c_3 + 6ab) - q^2(3s^2 ab + 6sa^3) + q^3(2s^3 + 16P_0)a^3. \end{aligned}$$

Put $e_1 = sa$, $v_1 = sb + 2a^2$, $r_1 = sc_3 + 6ab$. The net raw mean change is exactly $\varepsilon e_1 q$. For the total variance and additive third-moment quantity,

$$\begin{aligned} V(q) - 1 &= q(v_1 + O(\epsilon h^2)) - q^2(e_1^2 + O(\epsilon h^2/\lambda)), \\ \mathcal{B}'(0) &= r_1 + O(\epsilon h^3), \\ \mathcal{B}(q) - \beta - q\mathcal{B}'(0) &= -3e_1 v_1 q^2 + (2s^3 + 16P_0)a^3 q^3 + O(\epsilon q^2 h^3/\lambda). \end{aligned}$$

For the old coordinates these errors follow by differentiating their affine means: their summed second cubic derivatives are at most $K\lambda^{-2}\sum r_i^2 h^3 \le K\epsilon h^3/\lambda$. The term $16P_0 a^3 q^3$ is retained as the cost of centering the new coordinate.

Evaluate at centered threshold $z - \varepsilon e_1 q$ so that the raw event stays fixed. Every old first score is exactly zero. The fresh first derivative is the insertion of two marks, hence $J'(0) = -P_2$, where

$$P_2 = \iint d(\varepsilon(x + y)) \, d\mu(x) d\mu(y) \le K\epsilon E_m(h)/\lambda.$$

Lemma 11.1 bounds distinct-coordinate selections; selections from the same original coordinate have weight at most $\sum r_i^2/\lambda^2 \le \epsilon/\lambda$ and use the envelope. The lower bound below retains the resulting term $-qP_2$.

For completeness, the smooth normalization after removing the polynomial part of the CDF is

$$\mathcal{N}(e,v) = (1+v)^{3/2}[F(z) - \phi e - A(v+e^2) - \Phi((z-e)/\sqrt{1+v})] - D.$$

Its gradient at zero vanishes and its Hessian has entries $-3D, \phi''(z)/2, -L_0$. Combining the preceding exact moments, (11.4), and the factor $(1+v)^{3/2}$ on the remaining cubic term gives

$$J(q) \geq -qP_2 + \frac{q^2}{2}\{-3De_1^2 + 2c_\varepsilon e_1 v_1 - L_0 v_1^2 + 3Cv_1 r_1\} \\ -K\{\epsilon q^2 E_m(h)/\lambda + q^3 h^3\}.$$

The product term involving $v_1 P_2$ is absorbed in the displayed error. The quadratic third-moment term supplies $6Ce_1v_1$ in the Hessian; together with the normal mixed term this is $2c_\varepsilon e_1 v_1$. All remaining smooth terms are covered by q^3h^3 or $\epsilon q^2 E_m(h)/\lambda$, since $E_m(h) \geq h^3$. Now $e_1 = c_\varepsilon(2a^2)/(3D)$, $v_1 = 2a^2 + O(h^3)$, and $r_1 = O(h^3)$. The braces are $4a^4(c_\varepsilon^2/(3D) - L_0) + O(h^5) \geq ch^4$. Choose $q = c_0 \min(\lambda, h)$ with a small fixed c_0. The centering cost is absorbed and $J(q) \leq 0$ proves (11.2). □

12 From grouping rates to weighted summability

Lemma 12.1 (A grouping rate implies weighted variance summability)**.** *Suppose that after deletion of any selected set of original coordinates, the remaining coordinates can, for every large J, be partitioned into at most KJ groups with cost $w \leq KJ^{-1-\theta}$, where $\theta > 0$ is fixed and the constant is independent of the selected set. Then*

$$V(h) = o(h^p) \quad \left(1 < p < \frac{1+2\theta}{1+\theta}\right), \qquad W_s < \infty \quad \left(0 < s < \frac{1+2\theta}{1+\theta}\right).$$

Proof. If $V(h) \geq ch^p$ along a sequence, select a finite original block with radii at most h and variance $\sigma^2 \asymp h^p$, by stopping when a fixed smaller multiple of h^p is reached. Its variance overshoot is at most $h^2 = o(h^p)$. Its independent remainder is denoted by A_0 and has CDF G. The majorant in Lemma 9.2 has bounded coefficients and $b + \phi(z) = o(1)$, while Lemma 9.3 gives $\mathbb{E}Q_U(\sigma Z) = O(\sigma^3 + h) = O(h)$.

Choose fixed exponents

$$\frac{p(1+\theta)}{1+2\theta} < \lambda < 1, \qquad \frac{\lambda}{1+\theta} < j < 2\lambda - p, \qquad \ell = h^\lambda, \quad J = \lceil h^{-j} \rceil.$$

All intervals are nonempty. In particular $\ell = o(\sigma)$, $N\ell^2 \log(1/h) = o(\sigma^2)$, and $w, h, \sigma^3 = o(\ell)$. Group the unselected remainder, append one ordinary $N(0,\sigma^2)$ coordinate, and phase-round these $N+1$ coordinates with total phase z as in (9.10). If W_N is the sum of N independent triangular noises, that identity gives exactly

$$\mathbb{E}_{\text{phase}} F_{\text{rounded}}(z) = \mathbb{E}_{0 \leq u \leq \ell} F_{A_0 + \sigma Z + W_N}(z+u).$$

The right-hand average is uniform in u.

We check the positive slack under this additional noise. Hoeffding's elementary exponential-moment bound gives $\mathbb{E}e^{tW_N} \leq e^{N\ell^2 t^2/2}$. If g_u is the density of $\sigma Z + W_N - u$, then

$$\frac{g_u(\sigma x)}{\phi_\sigma(\sigma x)} \leq \exp\{\ell|x|/\sigma + N\ell^2 x^2/(2\sigma^2)\} \leq 2$$

on $|x| \le \sqrt{160\log(1/h)}$, uniformly for $0 \le u \le \ell$. The tails cost $o(h)$ against $Q_U \le K(1+|u|^3)$ by the same subgaussian bound and sixth moments. Hence $\sup_{0\le u\le \ell} \mathbb{E}Q_U(\sigma Z + W_N - u) = O(h)$. For $R_u = \sigma Z + W_N - u$, the exact polynomial identity is

$$F_{A_0+\sigma Z+W_N}(z+u) - F(z) = -bu + c(N\ell^2/6 + u^2) + C\{\mathbb{E}|R_u|^3 - \mathbb{E}|U|^3\} + \mathbb{E}Q_U(U) - \mathbb{E}Q_U(R_u).$$

Its last positive term may be retained as zero in a lower bound. Every term apart from $-bu$ is bounded below by $-K(h + \sigma^3 + N\ell^2 + \ell^2) = o(\ell)$. The phase-averaged CDF gain is thus $\phi(z)\ell/2 - o(\ell)$.

For every phase the variance lies between 1 and $1 + (N+1)\ell^2/4 = 1 + o(\ell)$. Its third-moment sum is at most

$$\beta + w + K\sigma^3 + K\ell^2(\sqrt{N} + \sigma) + K(N+1)\ell^3 = \beta + o(\ell).$$

The removed third-moment sum was nonnegative; the upper bound charges the new Gaussian and every grouped and rounded coordinate. Normalization costs $o(\ell)$ uniformly. Some phase therefore has positive objective, a contradiction. Finally (10.1), with any larger available exponent $p > s$, proves the assertion about W_s. □

Lemma 12.2 (An initial polynomial grouping rate)**.** *Under the hypotheses of Lemma 8.2, the uniform grouping rate in Lemma 12.1 holds with $\theta = 1/80$. Consequently $W_s < \infty$ for every $s < 82/81$ and $\sum_i \mathbb{E}|X_i| < \infty$.*

Proof. We have $W_{3/4} < \infty$ by Lemma 10.1. Among coordinates with $v_i \le \delta$ and $H_i > r_\delta = \delta^{1/20}$, balanced extremes force a triple's middle state to have modal deficit $\theta_i \le K\delta/H_i^2$. Indeed the extreme contributions to variance bound their total probability, and the centered middle state has magnitude at most $\theta_i H_i$. All its nonmodal raw gaps are therefore comparable to H_i. For two-point laws, $v_i = \theta_i(1-\theta_i)H_i^2$ gives the same conclusion. The finitely many exceptional coordinates are absent for small δ.

In each dyadic shell $h < H_i \le 2h$, apply Lemma 11.2 separately to the two physical signs, with $\epsilon \le K\delta/h^2$ and $E_m(h) \le K$. It gives $\lambda \le K(\sqrt{\delta}\, h^{-3} + \delta h^{-7})$. For $h \ge r_\delta/2$ this tends to zero uniformly. If a sign's total intensity reached one, select fractions totaling one to obtain a contradiction. The entire sign intensity is therefore less than one and obeys the same bound. Since $v_i \le \theta_i H_i^2$, summing shells gives

$$\mathcal{W}(\delta) := \sum_{v_i \le \delta} v_i/\sqrt{H_i} \le W_{3/4} r_\delta^{1/4} + K(\sqrt{\delta}\, r_\delta^{-3/2} + \delta r_\delta^{-11/2}) \le K\delta^{1/80}.$$

Now remove any selected set. Retain all remaining $v_i > J^{-1/2}$ individually, at most $\sqrt{J}$ of them. At the fine cutoff $r = J^{-2/3}$, the bounds from the full representation are $M(r) \le KW_{3/4} r^{-1/4}$ and $V(r) \le W_{3/4} r^{3/4}$. The remaining annular class sum is at most $K\mathcal{W}(J^{-1/2})$. Annular and variance grouping give $O(J)$ groups and

$$Jw \le K\{M(r)^3/J + \mathcal{W}(J^{-1/2})^2 + V(r)^{3/2}\sqrt{J}\} \le K(J^{-1/2} + J^{-1/80} + J^{-1/4}).$$

These bounds also hold after deleting coordinates, so the constant is uniform over selected sets. Lemma 12.1 proves the weighted assertion, and (8.2) with any $s > 1$ gives first-moment summability. □

Lemma 12.3 (The envelope on a dyadic diameter interval)**.** *Suppose $W_s < \infty$ for some $s > 1$. If a selected block from $h < H_i \le 2h$ has variance $\sigma^2 \ge ch^{2-\eta}$, where $\eta > 0$ is fixed, then $E_m(h) \le Kh$ for every fixed $m > \max(3, 2/\eta)$.*

Proof. Its coordinate radii are at most $2h$, $h/\sigma = O(h^{\eta/2})$, and $\sigma^2 \le (2h)^s W_s = o(h)$. The majorant has $b + \phi(z), c + A = O(\sigma)$ and $\mathbb{E}Q_U(\sigma Z) = O(h)$. Set $L = \sqrt{160 \log(1/h)}$, $r_* = \sigma/L$. For $|t| \le r_*$, the Gaussian density ratio on $|u| \le \sigma L$ is bounded by $e^{1+1/(2L^2)}$, and its cubic tails cost $o(h)$. The weighted Stieltjes estimate in Lemma 9.3, applied to $Q_U(\sigma x + t)$, then gives

$$\sup_{|t| \le r_*} \mathbb{E}Q_U(U + t) \le Kh.$$

Its error is bounded by $K(h/\sigma)\{\sigma + (1 + L^3)\mathbb{E}Q_U(\sigma Z + t) + (1 + L^2)\phi(L)\}$, with both endpoints included; the actual tail is $o(h)$ by Bernstein and sixth moments. The exact averaged polynomial identity therefore gives $0 \le d(t) \le K(h + \sigma^2 + \sigma^3) \le Kh$ for $|t| \le r_*$. For larger radii use $d(t) \le K(1 + |t|^3)$:

$$\sup_{r \ge r_*} (h/r)^m \sup_{|u| \le r} d(u) \le K(h/r_*)^m \le Kh^{m\eta/2} L^m = o(h).$$

Together with the added h^3 in (11.1), this proves the claim for the full envelope. □

Lemma 12.4 (Coordinates with small variance or small $p_i H_i$)**.** *If $1 < s < 12/7$ and $W_s < \infty$, then for $q = 0, 1/2$,*

$$\sum_{v_i \le \delta} v_i H_i^{-q} \le K_s \delta^{(s-q)/6}. \tag{12.1}$$

If all sufficiently late coordinates are two-point, write them $X_i = \varepsilon_i H_i (B_{p_i} - p_i)$ with $p_i \le 1/2$. For $1 < s < 11/6$ with $W_s < \infty$,

$$\sum_{p_i H_i \le \delta} p_i H_i \le K_s \delta^{(s-1)/5}. \tag{12.2}$$

Proof. For the first assertion choose $0 < \eta < 2 - 7s/6$ and then $m > \max(3, 2/\eta)$. In the variance sublevel, each shell above $r_0 = L_0' \delta^{1/7}$ has modal deficit at most $K\delta/h^2$. The global intensity bound from the preceding proof applies. Choose the fixed constant L_0' large enough that its right side is below one at $h = r_0/2$. It bounds the full shell intensity above that cutoff.

If a shell has total nonmodal intensity at least $h^{-\eta}$, its actual variance is at least $ch^{2-\eta}$: the mode has mass bounded below, the selected gaps are comparable to h, and $\operatorname{Var} Y = \frac{1}{2}\mathbb{E}(Y - Y')^2$ includes the mode–nonmode pairs. Lemma 12.3 gives $E_m(h) \le Kh$. Selecting one unit of intensity in a sign and using (11.2) forces $\delta \ge ch^6$. Thus above $r_1 = L_1' \delta^{1/6}$ the shell intensity is less than $h^{-\eta}$, with a sufficiently large fixed L_1'. For small δ, $r_1 < r_0$. The low, middle and high ranges have the following bounds:

$H_i \le r_1$	$K\delta^{(s-q)/6}$
$r_1 < H_i \le r_0$	$K\delta^{(2-q-\eta)/7}$
$H_i > r_0,\ v_i \le \delta$	$K\{\delta^{(5-2q)/14} + \delta^{(2-q)/7}\}$.

The low range uses W_s; the others sum geometric shell bounds h^{2-q} times their actual intensities. The chosen range of s, η makes the last two rows no larger than the first, proving (12.1).

For a two-point coordinate put $m_i = p_i H_i$. On $m_i \le \delta$, the modal deficit is $p_i \le \delta/h$ in a shell. The global intensity bound is $K(\sqrt{\delta}\, h^{-5/2} + \delta h^{-6})$, valid for the entire shell above $r_0 = L_0' \delta^{1/6}$ by the same unit-selection argument. Choose $0 < \eta < (11 - 6s)/5$. If the shell intensity is at least $h^{-\eta}$, then Lemma 12.3 gives $E_m(h) \le Kh$, and selecting one unit of intensity forces $\delta \ge ch^5$. Thus shell intensity is below $h^{-\eta}$ above $r_1 = L_1' \delta^{1/5}$. The three contributions to $\sum m_i$ are

$$K\delta^{(s-1)/5}, \qquad K\delta^{(1-\eta)/6}, \qquad K(\delta^{1/4} + \delta^{1/6}).$$

For the first use $m_i = v_i/((1-p_i)H_i) \le 2v_i/H_i$. The stated range of s, η makes all exponents at least $(s-1)/5$. The finite exceptional prefix is absent from either sublevel for sufficiently small δ. □

Proof of Theorem 1.3. The elementary counting implication

$$\sum_{a_i \le \delta} a_i \le K\delta^b, \quad 0 < b < 1 \quad \Longrightarrow \quad \#\{i : a_i > \delta\} \le K'\delta^{b-1}$$

follows by summing the dyadic intervals $(2^k\delta, 2^{k+1}\delta]$. For the three-point representation apply it to (12.1) with $q = 0$ and choose $\delta = J^{-6/(6-s)}$. Retain all $v_i > \delta$, at most KJ original coordinates. The remaining $W_{1/2}$ is at most $KJ^{-(s-1/2)/(6-s)}$. Since the full first-moment sum is finite, annular grouping above diameter J^{-1} and variance grouping below it give, after any selected deletion,

$$w \le K_s\{J^{-2} + J^{-1-2(s-1/2)/(6-s)} + J^{-1/2-3s/2}\} \le K_s J^{-1-\theta_s}, \qquad \theta_s = \frac{2s-1}{6-s}.$$

The class count is $O(\log J)$ and is included in $O(J)$. Lemma 12.1 therefore sends every available $s \in (1, 12/7)$ to every $\gamma < T(s) = (3s+4)/(s+5)$. Initially an $s > 1$ is available by Lemma 12.2. Availability is downward closed. If its supremum below $\sqrt{5}-1$ were s_*, choose available s close enough to s_* that $T(s) > s_*$, a contradiction. Here $T(s) > s$ exactly below the positive fixed point $\sqrt{5}-1$. Every use fixes s before the limit; no uniform constants or conclusion at the endpoint are needed.

For the two-point tail apply the counting implication to (12.2) and choose $\delta = J^{-5/(6-s)}$. The retained count is $O(J)$ and the remaining endpoint mean is at most $KJ^{-(s-1)/(6-s)}$. Group the coordinates according to the sign of their nonmodal displacement. Within each sign class, apply the grouping estimate after reflection if necessary, using the nonnegative raw variables $H_i B_{p_i}$. Their negative first-moment term is zero. Lemma 9.1 gives

$$w \le K_s J^{-2-3(s-1)/(6-s)} = K_s J^{-1-\theta_s^{\mathrm{B}}}, \qquad \theta_s^{\mathrm{B}} = \frac{2s+3}{6-s}.$$

The finite original prefix is retained individually. This is uniform after any selected deletion. The new map is $T_{\mathrm{B}}(s) = (3s+12)/(s+9)$, with positive fixed point $\sqrt{21}-3 < 11/6$. Starting from any available $s > 1$, the same supremum argument proves the improved range. First-moment summability already followed from Lemma 12.2. □

12.1 Regularity at the selected threshold

Proof of Corollary 1.4. The general grouping rate above has some $\theta > 0$. If $F(z+t) = F(z) + \phi(z)t + O(t^2)$ through both signs, take $\ell = J^{-a}$ with $1 < a < 1+\theta$. The exact phase kernel (9.10) has average linear gain $\phi(z)\ell/2$ and quadratic error $O(J\ell^2) = o(\ell)$. Concentration pays its excursion outside the stated neighborhood. Its actual grouping, rounding and normalization costs are $o(\ell)$, as in the preceding phase proofs. This contradicts maximality.

More generally, suppose only the left expansion $F(z) - F(z-h) = \phi(z)h + O(h^{1+\alpha})$ holds. It is enough to take $0 < \alpha \le 1$, decreasing a larger exponent if necessary. A grouping rate $w = O(J^{-1-\theta})$ excludes it whenever $\alpha > 1/(1+2\theta)$. Choose

$$\frac{1+\alpha}{2\alpha} < a < 1+\theta, \qquad \ell = J^{-a}.$$

Evaluate the rounded array at the lattice point $z - b_J\ell$, where the integer $b_J = O(\sqrt{J \log J})$ is large enough that $-b_J\ell - W_\ell \in [-2b_J\ell, 0]$ except with probability $O(J^{-L})$, for a fixed $L > a + 1$. Hoeffding's inequality supplies this event. The exact noise mean and the left expansion give

$$\mathbb{E}_{\text{phase}} F_{\text{rounded}}(z - b_J\ell) = F(z) - \phi(z) b_J \ell + \phi(z)\ell/2 + o(\ell),$$

since $\ell^\alpha (J \log J)^{(1+\alpha)/2} \to 0$. The bounded CDF pays the exceptional event. Normal Taylor error is $O(J\ell^2 \log J) = o(\ell)$; the variance cost is $O(J\ell^2)$, and the actual cubic cost is $w + O(\sqrt{J}\ell^2 + J\ell^3) = o(\ell)$. Thus the same positive continuity correction remains.

For the two-point tail, the rate just proved has $\theta = (2s + 3)/(6 - s)$ for every fixed available $s < \sqrt{21} - 3$. Its threshold is $1/(1 + 2\theta) = (6 - s)/(3s + 12)$, whose limit is $(\sqrt{21} - 3)/6$. Choose s sufficiently close to the endpoint. □

13 Restrictions on infinite Bernoulli tails

The contact equations also restrict the spacing of infinitely many two-point coordinates. The orientation matters: throughout this section a selected coordinate is

$$X_j = h_j(p_j - B_j), \qquad B_j \sim \text{Bern}(p_j), \quad h_j > 0.$$

Its negative value has probability p_j. We assume $p_j \to 0$. All statements concern a complete maximizing representation and the threshold in (1.5). We do not assume that a conditional or deleted law is maximizing.

13.1 A probability equation at the maximizing threshold

Put $\varphi = \phi(z)$ and

$$d(t) = F(z) + \varphi t - At^2 + Ct^3 - F(z + t), \qquad t \geq 0. \tag{13.1}$$

The Poisson contact inequality (3.3) gives $d \geq 0$. Monotonicity of F gives, with a fixed finite K,

$$d(t) \leq \varphi t + |A| t^2 + Ct^3, \qquad d(y) - d(x) \leq K(y - x) \quad (0 \leq x \leq y \leq 1). \tag{13.2}$$

For one selected coordinate put $p = p_j$, $q = 1 - p$, $h = h_j$. If $T_h f(t) = f(t + h)$, the distribution functions of S and $S - X_j$ satisfy the exact identity

$$\psi = (qI + pT_h)g, \qquad \psi(t) = F(z + t), \quad g(t) = F_{S - X_j}(z - ph + t).$$

For $p < 1/2$, the inverse is its convergent Neumann series. Thus, for a polynomially bounded function f with $f(0) = 0$,

$$[(T_h - I)(qI + pT_h)^{-1} f](0) = q^{-2} \sum_{k \geq 1} (-p/q)^{k-1} f(kh). \tag{13.3}$$

All series here converge absolutely. Varying the probability at the two fixed raw support points and applying (3.1) gives

$$[(T_h - I)(qI + pT_h)^{-1} \psi](0) = \varphi h - A(1 - 2p)h^2 + C(1 - 2p)^3 h^3.$$

The derivatives of raw mean, central variance and central absolute third moment are $-h$, $(1 - 2p)h^2$ and $(1 - 2p)^3 h^3$. This calculation uses probability derivatives, so it remains valid when z is an atom.

The same operator sends t, t^2, t^3 at zero to $h, (1-2p)h^2, (1-6pq)h^3$, respectively. Substituting (13.1) therefore yields

$$\sum_{k\geq 1}(-p/q)^{k-1}d(kh) = -2Cp^2q^2(3-4p)h^3 < 0. \tag{13.4}$$

In particular,

$$d(h) \leq \sum_{k\geq 2}(p/q)^{k-1}d(kh) \leq K_1 ph \quad (p \leq 1/4, h \leq 1), \tag{13.5}$$

where K_1 is uniform over the selected tail. The last bound follows by summing the linear, quadratic and cubic bounds in (13.2).

13.2 Successive gaps and right-hand regularity

Theorem 13.1. *Suppose a maximizing representation contains distinct original coordinates $h_j(p_j - B_j)$ with $p_j \to 0$, and h_j strictly decreasing to zero. If, for some $\nu > 1$,*

$$h_j - h_{j+1} = O(h_j^\nu), \tag{13.6}$$

then

$$F(z+t) = F(z) + \phi(z)t + O(t^{1+\alpha}), \qquad \alpha = \min(1, \nu - 1), \quad t \downarrow 0.$$

Under the hypotheses of Theorem 1.3, the representation is consequently impossible if

$$\nu > 1 + (\sqrt{5} - 1)/2.$$

Under the same hypotheses, if the complete representation is eventually two-point, the sufficient condition improves to

$$\nu > 1 + (\sqrt{21} - 3)/6.$$

The selected coordinates need not be the whole tail for the regularity conclusion. The exclusion uses the grouping estimates for the complete representation.

Proof. First take $1 < \nu \leq 3$. For each small $x > 0$, let $s(x)$ be the largest selected gap not exceeding x. Equation (13.6) gives $0 \leq x - s(x) \leq K_0 x^\nu$. Put $H(x) = \sup_{0\leq t\leq x} d(t)$. Choose $r_0 > 0$ so small that

$$W_\nu = \sum_{k\geq 2} r_0^{k-1}k^\nu < 1, \qquad W_3 = \sum_{k\geq 2} r_0^{k-1}k^3 < 1.$$

Eventually $p_j/(1-p_j) \leq r_0$. Combining (13.5) with the one-sided increment bound in (13.2), and taking a supremum, gives

$$H(x) \leq c_0 x^\nu + \sum_{k\geq 2} r_0^{k-1} H(kx) \quad (0 < x < \delta).$$

For a sufficiently large constant K_2, the function $B(x) = K_2(x^\nu + x^3)$ dominates H on $[\delta, \infty)$ and satisfies

$$B(x) - \sum_{k\geq 2} r_0^{k-1} B(kx) \geq c_0 x^\nu.$$

The positive part $v = (H - B)_+$ is bounded and vanishes outside $(0, \delta)$. It obeys $v(x) \leq \sum_{k\geq 2} r_0^{k-1} v(kx)$ for every $x > 0$. The sum of the coefficients is less than one, so taking the supremum forces $v = 0$. Hence $d(t) = O(t^\nu)$ at zero. Together with (13.1), this gives the asserted regularity. For $\nu > 3$, apply the argument with exponent three.

We record why a right-hand expansion suffices for the exclusion. Suppose the complete representation can be partitioned into $N = O(J)$ independent groups, increasing its additive third absolute moment by at most $w = O(J^{-1-\theta})$, with $\theta > 0$. A right-hand expansion of order $1+\alpha$, $0 < \alpha \le 1$, is incompatible with maximality when $\alpha > 1/(1+2\theta)$. Choose

$$\frac{1+\alpha}{2\alpha} < a < 1+\theta, \qquad \ell = J^{-a}.$$

Use the rounding phases of (9.10) and a lattice threshold $z + b\ell$, where b is a positive integer of order $\sqrt{J \log J}$ with a sufficiently large fixed coefficient. Bounded-range concentration ensures $0 \le b\ell - W_\ell \le 2b\ell$ except with probability $O(J^{-L})$ for any chosen fixed $L > a+1$. The right-hand expansion and $\mathbb{E}W_\ell = -\ell/2$ give

$$\mathbb{E}_{\boldsymbol{u}} F_{L_{\boldsymbol{u}}}(z + b\ell) = F(z) + \varphi b\ell + \varphi\ell/2 + o(\ell).$$

The remainder is $O((\sqrt{J \log J}\,\ell)^{1+\alpha}) = o(\ell)$. The exceptional probability and first moment cost are $o(\ell)$ as well. The normal comparison contributes $\Phi(z) + \varphi b\ell + O(J\ell^2 \log J)$. The actual variance change is $O(J\ell^2)$; grouping and rounding increase the cubic quantity by at most

$$w + O(\sqrt{J}\,\ell^2 + J\ell^3) = o(\ell).$$

All normalization errors are $o(\ell)$. The positive average gain $\varphi\ell/2 + o(\ell)$ contradicts maximality at the comparison threshold.

The grouping estimates for the whole representation proved in Section 9 and the proof of Theorem 1.3 supply

$$\theta = (2s-1)/(6-s), \quad 1 < s < \sqrt{5} - 1,$$

and, for an eventually two-point representation,

$$\theta = (2s+3)/(6-s), \quad 1 < s < \sqrt{21} - 3.$$

Inserting these in $\alpha > 1/(1+2\theta)$ and taking s sufficiently close to its upper endpoint proves the two stated strict thresholds. □

For actual gaps $h_j = aj^{-r}$ at every sufficiently large index, (13.6) holds with $\nu = 1 + 1/r$. Thus an admissible complete maximizing representation with $\beta \le 18481/25000$ is excluded for $0 < r < (\sqrt{5}+1)/2$, or for $0 < r < (\sqrt{21}+3)/2$ if it is eventually two-point. An asymptotic equivalent for h_j alone does not control its successive increments.

13.3 Nonoverlapping support intervals

Theorem 13.2. *A maximizing representation cannot have the form*

$$S = P + \sum_{j \ge 1} h_j (p_j - B_j),$$

where P has finite support and is independent of the tail, $0 < p_j \le 1/2$, $p_j \to 0$, and, for some fixed $c > 0$,

$$T_j := \sum_{k > j} h_k \le h_j, \qquad h_{j+1} \ge c h_j > 0$$

at every sufficiently large index. In particular, this excludes geometric gaps $h_j = ar^j$ with $0 < r \le 1/2$ and $p_j \to 0$. No bound on β beyond admissibility is required.

Proof. Absorb a finite initial segment into P, and suppose the gap conditions hold at every index. Put $M(t) = F(z+t) - F(z)$. Equations (13.1)–(13.5) give

$$M(t) \le K_3 t \quad (t > 0), \qquad M(h_j) \ge \varphi h_j - |A| h_j^2 - K_1 p_j h_j.$$

The gaps decrease, and $h_j \ge h_{j+1} + h_{j+2} \ge 2h_{j+2}$. Consequently, for all sufficiently large j,

$$M(h_j) - M(h_{j+2}) \ge \varphi h_j / 3. \tag{13.7}$$

Write $b = \sum_j p_j h_j$ and $K = \{\sum_j h_j \varepsilon_j : \varepsilon_j \in \{0,1\}\}$. This compact set is the support of $\sum_j h_j B_j$. For each support state a of P with mass w_a, put $y_a = a + b - z$. States with $y_a \notin K$ contribute nothing to sufficiently small right-hand intervals at z. For each remaining state choose one expansion $y_a = \sum_j h_j \varepsilon_{a,j}$ and put

$$v_{a,j} = w_a \prod_{k<j} p_k^{\varepsilon_{a,k}} (1-p_k)^{1-\varepsilon_{a,k}}.$$

If $\varepsilon_{a,j} = 1$, its modal child lies at a displacement from z between zero and $2h_j$. A zero displacement would require an infinite all-one tail, whose probability is zero. Hence

$$v_{a,j}(1-p_j) \le M(2h_j) \le 2K_3 h_j, \qquad v_{a,j} \le 4K_3 h_j. \tag{13.8}$$

Distinct expansions of y_a can only meet at a support-interval endpoint when $h_j = T_j$. Such endpoints do not lie in the strict raw interval $[y_a - t, y_a)$ corresponding to the right-hand increment of the closed CDF.

For a point in $[y_a - h_i, y_a)$ whose first different digit is at $j < i$, the chosen digit must be one and the new digit zero. The new tail must be within h_i of its maximum T_j. Thus its digits $j+1, \dots, i$ are all one, almost surely; the only exception would again require an infinite all-one continuation. Its first-difference class has mass at most $v_{a,j}(1-p_j)\prod_{k=j+1}^{i} p_k$. Equation (13.8) and $h_j/h_i \le c^{-(i-j)}$ bound the total of these earlier classes by

$$4K_3 h_i R_i, \qquad R_i = \sum_{j<i} \prod_{k=j+1}^{i} (p_k/c).$$

The recurrence $R_{i+1} = (p_{i+1}/c)(1+R_i)$ gives $R_i \to 0$. There are only finitely many states of P, so all earlier classes together have mass $o(h_i)$.

A first different digit after $i+2$ has displacement at most $T_{i+2} \le h_{i+2}$ and cannot contribute to the annulus in (13.7). For large i, at least $\varphi h_i/4$ therefore comes from indices $i, i+1, i+2$. If P has d states, one pair (a,j) among these $3d$ possibilities satisfies

$$\varepsilon_{a,j} = 1, \qquad v_{a,j} \ge c_1 h_j, \qquad c_1 = \varphi/(12d) > 0.$$

Once this occurs at j, every later prefix of that component includes the rare factor p_j. For fixed L and $j < k \le j + L$,

$$\frac{v_{a,k}}{h_k} \le 4K_3 c^{-L} p_j \longrightarrow 0.$$

The same component cannot supply two such pairs within L late indices. Choose $L = 3d+3$ and $d+1$ disjoint three-index windows starting at $N, N+3, \dots, N+3d$. They require $d+1$ pairs among d components over a span smaller than L, a contradiction. □

A Rational checks for the Gaussian Fourier bound

The finite calculation in Lemma 5.3 uses only rational arithmetic. For $x \geq 0$ define

$$E_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}, \qquad U_n(x) = E_n(x) + \frac{x^{n+1}}{(n+1)!} \frac{1}{1 - x/(n+2)} \quad (0 \leq x < n+2).$$

Then $E_n(x) \leq e^x \leq U_n(x)$, by the positive exponential series and a geometric bound on its remaining terms. The exponent $-t^2/2 + (269/5000)t^3$ is decreasing on $[0, 6]$. With $d = 3/100$, $l_j = jd$, $r_j = (j+1)d$, the exact upper sum

$$\sum_{j=0}^{199} \frac{d\, r_j^2}{E_{40}(l_j^2/2 - (269/5000) l_j^3)} < \frac{23}{10} \tag{A.1}$$

therefore bounds I. Integration by parts and the normal Mills bound give

$$J \leq \frac{6025}{216} e^{-108/25} \leq \frac{6025}{216 E_{40}(108/25)} < \frac{3}{8}.$$

The remaining rational comparisons are

$$\frac{U_{24}(25/54)}{6(157/50)} < \frac{17}{200}, \qquad \frac{11}{19 E_{40}(361/289)} < \frac{1}{6},$$
$$\frac{25}{61} + \frac{(3051/5000)(17/20)2}{(443/250)(9/10)(237/250)} < \frac{11}{10}.$$

The last expression bounds the Gaussian-convolution density because

$$(61/25)^2 < (19/10)(157/50), \quad (443/250)^2 < 157/50, \quad (237/250)^2 < 9/10.$$

The normal tail bound follows from $T > 7$, $\beta < 3/10$, $\pi > 3$, $c > 1$, $e > 5/2$, and $(5/2)^{24} > 10^8$. All these are finite rational inequalities, including (A.1); the executable companion `gaussian_window_certificate.py` evaluates them using Python's standard-library rational type and asserts each comparison. It contains no floating-point quadrature or external numerical library. Its final exact comparison is

$$\frac{1182125003}{3000000000} < \frac{2}{5}.$$

The classical numerical constant 0.3051 in (5.7) is an external theorem of Korolev and Shevtsova; this finite check verifies the additional comparisons used here.

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